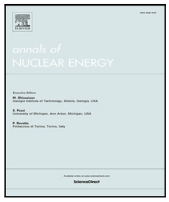
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Griffin: A MOOSE-based reactor physics application for multiphysics simulation of advanced nuclear reactors

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ABSTRACT

Griffin is a Multiphysics Object-Oriented Simulation Environment (MOOSE) based reactor physics application for multiphysics simulations of advanced reactor designs jointly developed by Idaho National Laboratory and Argonne National Laboratory. This paper summarizes the motivation, significance, architecture, design, and features of Griffin. Griffin offers flexible and extensible features to address the challenges associated with advanced reactor designs. These features range from fundamental particle transport to specific reactor physics tasks. The features cover a wide range including on-the-fly and traditional two-step cross-section generation methods, steady-state and transient transport solvers suitable for both heterogeneous and homogeneous models, high-fidelity depletion where thousands of isotopes can be tracked and low-fidelity depletion characterized by burnup, etc. The most fundamental aspect that sets Griffin apart from other reactor analysis codes is that it is developed based on the MOOSE framework. A modular development approach is strongly enforced, with multiphysics being an essential element considered since the beginning of Griffin's development. Griffin links various MOOSE physics modules and couples to other MOOSE-based applications and non-MOOSE-based applications for multiphyiscs simulations. Griffin includes three modules: ISOXML for preparing and managing multigroup cross sections, radiation transport for solving the neutron transport equation, and reactor analysis for user-oriented reactor physics analysis functionalities. Griffin uses various finite element methods for spatial discretization, multigroup approximation for energy discretization and discrete ordinates method, spherical harmonics expansion method, and diffusion approximation for streaming direction discretization to solve the neutron transport equation. Griffin's flexibility is evidenced through Griffin's various applications to fast reactor, high-temperature reactor, pebble bed reactor, molten salt reactor, and microreactor designs. Griffin development follows the software quality assurance procedure for MOOSE-based applications and with software requirements consistent with the ASME NQA-1 standard. Griffin has been adopted into the reactor analysis system for the U.S. NRC and is in use at U.S. companies, universities and national laboratories.

1. Motivation and significance

Griffin is a reactor physics application based on Multiphysics Object-Oriented Simulation Environment (MOOSE) (Lindsay et al., 2022; Giudicelli et al., 2024). Griffin is jointly developed by Idaho National Laboratory (INL) and Argonne National Laboratory (ANL) and is supported by the Nuclear Energy Advanced Modeling and Simulation (NEAMS) program administered by the Office of Nuclear Energy of the U.S. Department of Energy. Griffin development began in early 2020 (Martineau, 2021) as a combination of INL's MAMMOTH (DeHart et al., 2019) and Rattlesnake (Wang et al., 2021b, 2018b) codes while bringing in features of ANL's PROTEUS (Jung and Lee, 2018) code. The fast reactor multigroup cross-section generation code MC^2-3 (Lee and Yang, 2017) can be optionally linked.

Griffin is primarily developed for the design and analysis of various types of advanced nuclear reactors and test reactors. The integration

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of legacy codes developed by INL and ANL was necessary to rapidly deploy a reactor physics code with significant capability to tackle the large variety of reactor designs under deployment today. This allowed the focus of NEAMS resources to meet user needs, including industry and government organizations, like the U.S. NRC. Griffin inherits the framework-based design features presented in Ref. Wang et al. (2021b): flexibility, compatibility, extensibility, robustness with a Software Quality Assurance (SQA) procedure consistent with Nuclear Quality Assurance (NQA-1) requirements (Permann et al., 2020). The reliance on an unstructured mesh framework allows for a general geometric treatment, thus removing this significant limitation that plagues traditional nodal methods. All capabilities in Rattlesnake and MAMMOTH were incorporated into Griffin; although, some legacy solvers have been phased out. The fast-reactor cross-section preparation capability from MC^2-3 and the thermal reactor self-shielding capability from PROTEUS have also been incorporated into Griffin. Griffin has been deployed in the modeling and simulation of Fast Reactor (FR), High-Temperature Reactor (HTR), Pebble Bed Reactor (PBR), Molten Salt Reactor (MSR), and microreactor designs. A significant number of models using Griffin for reactor physics and coupled applications are available in the Virtual Test Bed (VTB) (Giudicelli et al., 2023), an open repository funded by the National Reactor Innovation Center (NRIC).

All nuclear reactors demonstrate simultaneous physical phenomena that are often tightly coupled in terms of their effects on each other (Duderstadt and Hamilton, 1976). This multiphysics nature can be more pronounced in the emerging advanced nuclear reactor designs. Fuel, structural and core geometries can significantly depart from traditional Light Water Reactor (LWR) designs. The reactor core structures may experience significant deformation during operation due to thermal expansion and radiation-induced dimensional changes. The neutron energy flux spectra can vary significantly in the evolving advanced reactor systems and ranges from fast, to epithermal and thermal due to the broad range of neutron moderating materials under consideration. For example, the use of TRi-structural ISOtropic (TRISO) fuel with graphite and metal hydride moderators can increase the neutron migration length and lead to significant spectrum variations in the active core regions. The flexibility and extensibility of the code is an essential feature since users will demand, as users historically have, more features and capabilities, including routine design calculations, reference calculations, support for uncertainty quantification, automated design optimization, online monitoring, load following, reactor maneuvers, etc.

Griffin was developed to meet these requirements through a modular approach based on the MOOSE framework. The MOOSE framework hosts features shared by all physics applications, including Griffin, such as:

- multidimensional high-order unstructured mesh support through libMesh libraries (Kirk et al., 2006)
- mesh partitioning and code parallelization
- a mesh generation system from the MOOSE reactor module (Shemon et al., 2023)
- a generic Preconditioned Jacobian-Free Newton-Krylov (PJFNK) solver (Knoll and Keyes, 2004) through Portable, Extensible Toolkit for Scientific Computation (PETSc) and Scalable Library for Eigenvalue Problem Computations (SLEPc) (Balay et al., 2015; Hernandez et al., 2005)
- MOOSE documentation system (MooseDocs) (Slaughter et al., 2021)
- consistent input/output formats across applications (Lefebvre et al., 2017)
- strongly and tightly multiphysics-coupling schemes (Gaston et al., 2015)
- framework-level standard yet versatile data transfers for multiphysics coupling
- controllable input parameters allowed to be changed during runtime, etc.

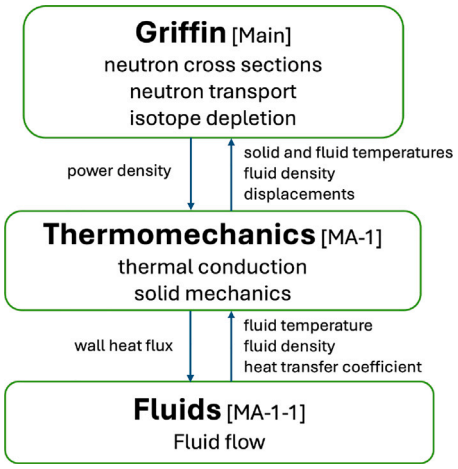


Fig. 1. Multiapp setup for a generic multiphysics simulation.

Griffin inherits various plug-and-play MOOSE physics modules, such as heat transfer, thermal hydraulics (Hansel et al., 2024), solid mechanics, solid and fluid properties. In addition, specialized modules like the reactor (Shemon et al., 2023), ray tracing (Gaston, 2020), stochastic tools (Slaughter et al., 2023), and optimization are also inherited from MOOSE. Griffin is deliberately modular in design with three components:

- ISOXML - the object-oriented multigroup cross section toolkit
- radiation transport - various transport discretization schemes and solvers
- reactor analysis - workflow for specific reactor designs, depletion, fuel management, etc.

Framework-based code development represents a major paradigm shift from the traditional approach, where the code is dedicated to a specific purpose and normally to a single reactor design. This framework-based development allows teams across applications (physics) and the framework to easily contribute features to the MOOSE source code that can be shared among applications, thus maximizing the code usage and minimizing the code duplication. We note that flexibility does not necessarily imply a lower performance than legacy code packages specifically developed for certain reactors, because the fundamental numerical techniques remain the same and are typically not significantly impacted by the choice of coding language or data structure.

Griffin can perform basic multiphysics simulations as a standalone code by relying on the MOOSE physics modules. For more advanced multiphysics simulations, Griffin is linked to other MOOSE-based applications including Pronghorn (Park et al., 2010), SAM (Hu, 2017), BISON (Williamson et al., 2012; Hales et al., 2016), Sockeye (Hansel et al., 2019), etc. and provided to costumers as a wrapped application such as with the U.S. NRC Comprehensive Reactor Analysis Bundle (BlueCRAB) or accessible via dynamic linking. A diagram of a typical multiphysics simulation configuration is depicted in Fig. 1, where Griffin is the main application coupled to a core thermomechanics (denoted with MA-1), which itself is coupled to a fluid solver (MA-1-1). The physics included in each sub-application are listed inside the boxes. The data transfers are listed by the arrows thus describing the flow of information. All applications are typically coupled via the Picard (fixed-point) iteration.

The stochastic tools and optimization modules facilitate uncertainty quantification and optimization of Griffin models. These tools are designed to directly interact with Griffin with efficient and scalable schemes for performing stochastic analysis in memory. Theory and user manuals are included in the documentation along with the code manual

generated by MooseDocs and Doxygen documentation. This documentation is published on the Griffin website and is updated with code changes. Continuous Integration, Verification, Enhancement, and Testing (CIVET), the building environment, and the software distribution provided by the MOOSE framework are used by all MOOSE-based applications, including Griffin. The INL's Nuclear Computational Resource Center (NCRC) provides multiple levels of Griffin licenses ranging from binary to source. Binary access is for typical users while source access is for advanced users that contribute to the code development effort.

This article lays out Griffin's software architecture and design, including the relation among the MOOSE framework, the MOOSE physics modules, and the three Griffin modules, followed by the details on software functionality. We first describe what capabilities MOOSE offers through its framework and modules. The description will be brief because most of these capabilities have been covered by Ref. Wang et al. (2021b). We then describe the three individual modules in Griffin. The discussion of the radiation transport module focuses on the development and improvements made for the Discontinuous Finite Element Methods (DFEM)-discrete ordinates method (SN) solver and the development for a new solver, Hybrid Finite Element Methods (HFEM)-spherical harmonics expansion method (PN). For ISOXML, we will introduce its design and describe the new features for supporting pin-power reconstruction, decay heat tracking with empirical standards or explicit isotope tracking, Bateman solver extensions, etc., and most importantly the Self-Shielding Application Programming Interface (SSAPI) (Lee and Jung, 2018b) for online cross-section generation, which was initially implemented for thermal reactor self-shielding and has been extended for TRISO-fueled reactors. We will describe the reactor analysis module, including the depletion system with decay heat tracking, fuel management, and workflow improvements that interact with ISOXML and MC^2-3, and the streamline-based modeling capability for PBR analyses. Griffin is under continuous development, therefore the specific features in this paper represent the current capabilities of the code. We will summarize the applications and impact of Griffin for various advanced reactor designs, emphasizing the multiphysics features of these applications. Some of these applications contain comparisons between modeling results and experiment measurements, which can be considered as validation points.

2. Software architecture/design

The most fundamental aspect that sets Griffin apart from most of the other reactor physics codes is that it is developed on a computational framework. Framework-based development helps the overarching features of Griffin. MOOSE is an open-source, finite-element, multiphysics framework written in C++ and is hosted by GitHub in a public git repository managed by INL. Its development is driven by the scientific community through both internal and external contributions. The MOOSE physics modules include software categorized into various physics and functionalities. Thermal conduction, fluid dynamics, solid mechanics, and meshing tools for reactor analyses are linked in Griffin. The ray tracing module (Gaston, 2020) built upon libMesh is used by the Griffin mesh sweeper. The stochastic tools module and optimization module are linked to support uncertainty quantification and optimization, which are often encountered during reactor design.

MOOSE also provides development infrastructure, including software compilation, regression test harness, continuous development integration, software distribution, etc. Griffin development follows the SQA procedure outlined by the MOOSE SQA planning document, the INL's PLN-4005 document (Permann et al., 2020). This plan is shared by all MOOSE-based applications and is consistent with American Society of Mechanical Engineers (ASME) NQA-1 specifications and requirements. The most important feature of this development procedure is the use of a common GitHub repository. MOOSE has a single-source documentation system based on fully customizable markdown syntax known as MOOSE style markdown, namely MooseDocs (Slaughter et al., 2021). The single-source markdown files may be converted into various media forms, including websites, slide shows, and PDFs (via LaTeX). More importantly, custom markdown syntax can be easily created, so it is possible to develop syntax to meet the needs of any project. MOOSE provides CIVET for automatic code development integration.

ISOXML is an object-oriented nuclear data toolkit independent of the rest of Griffin. It is the module that manages all nuclear data for reactor analyses. This data includes multigroup cross sections, isotope decay and transmutation data, homogenization equivalence data, form functions for fuel-assembly pin power reconstruction, empirical decay heat models, etc. More recently, we introduced SSAPI into ISOXML to support the online cross section generation capability for thermal reactors. ISOXML can use the transport solver provided by the radiation transport module to evaluate equivalent cross sections for online self-shielding calculations. SSAPI was enhanced to handle double heterogeneity for prismatic and pebble-bed graphite reactors with TRISO fuel particles. In addition, ISOXML includes fine-group thermal reactor cross-section library generation based on PyNJOY (École Polytechnique de Montréal, 2015), with NJOY2016 (Macfarlane et al., 2017) as a submodule.

The radiation transport module is a major part of Griffin. Although the module is mainly designed for reactor analysis with neutron, photon, or coupled neutron-photon calculations (referred to as neutronics), it supports transport calculations of other forms of radiation or particles, such as phonon and electron, to predict thermal conductivity (Harter et al., 2019) and thermal radiation for radiative heat transfer. The main goal of the radiation transport module is to perform neutron transport calculations that take a mesh, the multigroup cross-section libraries, and variables of other physics as inputs and provide the multiplication factor, the distribution of the neutron flux, various reaction rates, power density, etc. as outputs. This module supports neutronics calculations with spatially heterogeneous unstructured geometries for generating high-fidelity reference solutions and simulations with significant spatial homogenization for fast routine design calculations. It also provides the support required by the online cross-section generation interfacing SSAPI in ISOXML. The radiation transport module includes discretization schemes based on various Finite Element Methods (FEMs) on the unstructured mesh through the MOOSE framework. Particular solve techniques for these schemes, such as the transport mesh sweeping for DFEM-SN, the red-black iteration for HFEM-PN, etc., with the diffusion acceleration are part of this module. It is noted that these special solving techniques are necessary to obtain better performance than the general PJFNK solver available through the MOOSE framework. Transport materials interface with ISOXML in order to utilize the multigroup cross-section libraries, which allow for the evaluation of cross sections on local elements during transport calculations.

The reactor analysis module provides various functionalities required for reactor analysis. The radiation transport module focuses on solving the transport equations, while the reactor analysis module directly interfaces with users. Its functionalities include depletion calculations, criticality search, and post-processing tasks, such as pin-power reconstruction, fuel management, perturbation calculations, etc. Fission product poison tracking can be performed at saturation or with time-dependent poison tracking in depletion calculations. Time-dependent poison tracking during transient calculations is also supported in Griffin. The reactor analysis module also includes simplified workflows for specific reactor designs.

The Griffin software structure is illustrated in Fig. 2, including the major components of each module. Connecting arrows illustrate major software dependencies and should not be considered as the data dependency in a calculation. For example, ISOXML is an independent module that does not depend on MOOSE nor the other two Griffin modules. It can optionally use the transport solvers in the Griffin radiation transport module for the self-shielding treatment. ISOXML uses

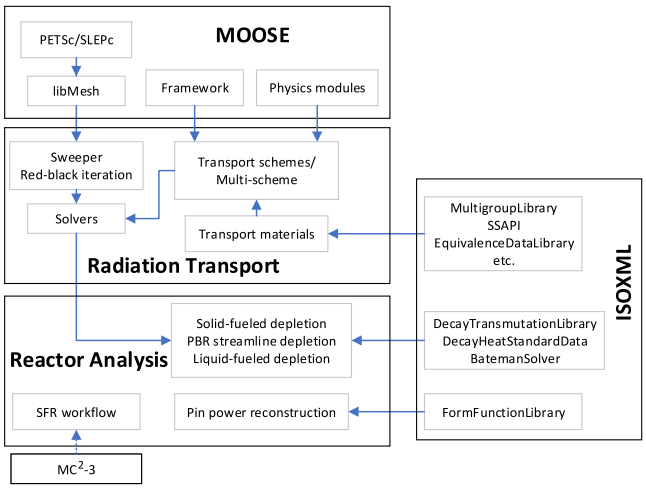


Fig. 2. An overall Griffin software structure.

markdown files for documentation that follow the MooseDocs extensions and can be processed by MooseDocs. The sweeper and red-black iteration in the radiation transport module depend directly on libMesh. The dependency of the sweeper on a task scheduler in the MOOSE ray-tracing module, which can be considered a libMesh extension, is not shown in the figure. ISOXML can be compiled in a standalone mode. Radiation transport can be compiled with MOOSE and ISOXML, and reactor analysis can be compiled with MOOSE, ISOXML, radiation transport, and optionally with MC^2-3.

3. Software functionality

We first describe MOOSE and its important features used by Griffin and then describe the functionality of the three modules of Griffin: ISOXML, radiation transport, and reactor analysis. We added new features in ISOXML and, most importantly, brought SSAPI into ISOXML to support online cross-section generation. We will briefly visit the multiphysics, multi-radiation, and multi-scheme features of the radiation transport module. We will detail the performance improvements to the DFEM-SN for heterogeneous high-fidelity simulations and the introduction of the new HFEM-PN scheme targeting homogeneous low-order models. For reactor analysis, we focus on our depletion feature for traditional solid fuel reactors with fuel management, a PBR streamline-based depletion capability with and without online cross-section generation, and a Sodium-cooled Fast Reactor (SFR) two-step workflow utilizing MC^2-3.

3.1. MOOSE framework and physics modules

3.1.1. General functionalities

MOOSE extends and incorporates two external libraries: libMesh (Kirk et al., 2006) and PETSc (Balay et al., 2015), which includes additional contributing packages. libMesh implements FEM capabilities on an unstructured mesh, while PETSc provides the algebraic nonlinear solvers and preconditioners for solving problems defined through MOOSE. Steady-state eigenvalue problems can be solved using solution algorithms in Scalable Library for Eigenvalue Problem Computations (SLEPc) (Hernandez et al., 2005), distributed along with PETSc. Capabilities implemented at the base of the MOOSE framework are available in all physics applications including Griffin, a distinct feature of framework-based development. Residual evaluations of the governing equations of various physics, including the neutron transport equation, can be implemented with MOOSE kernels, materials, auxiliary kernels, etc. MOOSE packages a broad range of common physics capabilities as MOOSE *physics modules*, including solid mechanics, thermal conduction, fluid dynamics, etc. MOOSE physics modules also include stochastic tools and optimization modules, which can leverage Griffin capabilities for creating reduced-order models, extending applications to uncertainty quantification, various design optimizations, and even reactor core monitoring and operational support. Multiphysics calculations can be achieved through a MultiApp/Transfer system in MOOSE for coupling multiple independently developed applications in a fully-coupled approach with fixed-point iterations or through a single system containing all physics solved by the PJFNK solver as a whole, referred to as the strongly-coupled approach. Multiphysics transient simulations can be restarted from a checkpoint file generated by a steady-state multiphysics calculation or a multiphysics depletion calculation at a time step. For a more complete description of MOOSE, we refer readers to Ref. Lindsay et al. (2022), Wang et al. (2021b).

3.1.2. Flexible iterative solving system in the MOOSE framework

MOOSE implements a flexible solving system that allows applications to create their own solve objects that can be integrated for a sophisticated iterative solving algorithm. A solve object is allowed to have an inner solve object optionally. For example with the transient solver or executioner in MOOSE, a time stepping/integration scheme is the outermost solver object. Its inner solve object is the multiphysics iteration solve object that iterates the solve of the main applications and sub-applications with MultiApps and data transfers. The inner solve object of the multiphysics iteration solve object calls the PETSc PJFNK solver. Examples of additional solve objects include mesh adaptation cycles, fixed point iterations, etc. Application developers are allowed to create their own solve objects and change the chain structure to achieve various iteration schemes. Custom execution flags are allowed to be added where various user objects, auxiliary kernels, etc. can be attached to and called by MOOSE for certain evaluations at certain execution points. We will exemplify this iterative solving system with the Griffin depletion solver with the DFEM-SN scheme in this paper.

3.1.3. Mesh generation

Griffin represents the geometry of a given problem using unstructured mesh capabilities inherited from libMesh. These capabilities include support for multiple dimensions (one-dimension (1D), two-dimension (2D), and three-dimension (3D)), mixed element types, curvilinear coordinates (RZ-cylindrical/R-spherical), first- and second-order iso-parametric elements, etc. These provide the flexibility in the geometric representation to address modeling and simulation challenges from various advanced reactor designs.

To facilitate the mesh generation, especially for various reactor geometries, MOOSE provides a mesh generator system in the MOOSE framework and MOOSE reactor module (Shemon et al., 2023). A mesh generation procedure is composed of multiple steps with ordered mesh generators, which can be as simple as renaming a mesh subdomain or stitching multiple mesh sections generated by input mesh generators. This mesh generator system significantly simplifies the reactor analysis workflow, reduces the efforts of mesh generation, and allows for a more straight-forward sensitivity analysis with geometry parameters. Users do not have to resort to other software for generating the mesh although the system does provide the capability for loading a pregenerated mesh. Mesh generation can be conducted on the fly employing hierarchical structures of reactors, ranging from pins and assemblies to reactor cores, with hexagonal or Cartesian configurations. The system is highly flexible and extensible, allowing new mesh generators to be easily integrated during the mesh generation steps. Information used for mesh generation, such as number of fuel assemblies, etc., labeled as mesh meta information, can be kept and retrieved later for cross-section generation, postprocessing, etc.

One unique feature in this mesh generation system is that arbitrary extra element integers can be added to a mesh. Elements of a mesh can have these element integers, which can be used by Griffin to assign

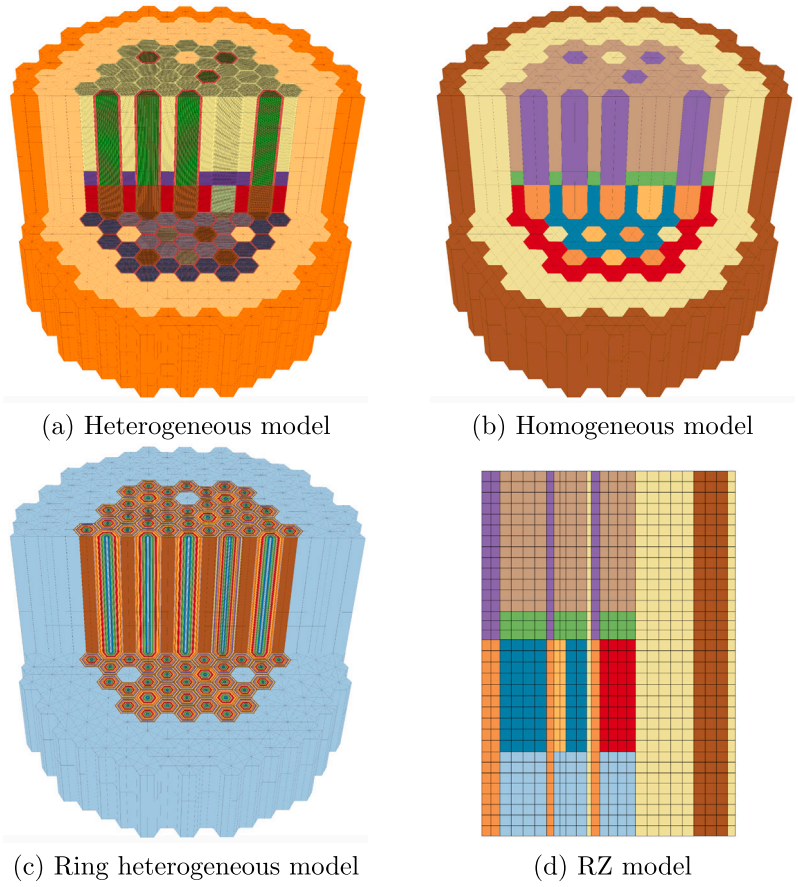


Fig. 3. Illustrations of the mesh-generation capability for a SFR model.

cross-section sets to elements and group elements together for homogenization equivalence, depletion calculations, postprocessing, etc. Consistency checks, such as ensuring that elements in one depletion zone are using the same cross-section sets, can be done with uniform framework interface functions.

As an illustration of mesh generation capabilities in Griffin, Fig. 3 shows the various models of Advanced Burner Test Reactor (ABTR) core designs (Terlizzi et al., 2024) with different levels of detail. Users can build the full-core model with the heterogeneous assembly details using the Reactor Geometry Mesh Builder (RGMB) provided by the MOOSE reactor module (Shemon et al., 2023). Then, the embedded mesh generator in Griffin, an equivalent core mesh generator, can convert this full-core model with the different details as illustrated in Fig. 3. It can significantly streamline the overall workflow for core analysis. This capability is extensively utilized in the fast reactor analysis described in Section 3.4.3. The mesh generated for the Empire reactor, a microreactor design, is shown in Fig. 4. The heterogeneous assemblies and control drums located in the core center and periphery regions, respectively, are explicitly meshed.

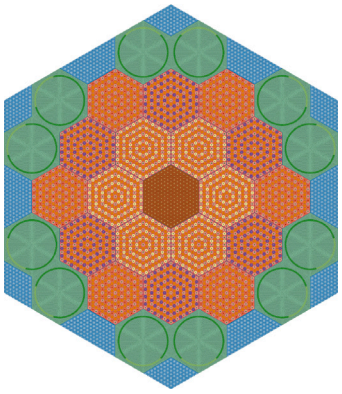


Fig. 4. Illustration of the heterogeneous mesh-generation capability for the Empire microreactor.

3.2. ISOXML

3.2.1. General description

The initial functionality of ISOXML was mainly to provide an Extensible Markup Language (XML) data format to store tabulated microscopic multigroup cross sections. These cross-section data are generated either with Monte Carlo codes or lattice physics codes, such as DRAGON (Marleau et al., 2011) or MC^2-3. The name, ISOXML, stems from ISOtopic XML, because of this functionality. ISOXML provides several data converters to process multigroup cross sections from Serpent (Leppänen, 2015), DRAGON, OpenMC (Romano et al., 2015) outputs, or ISOTXS files generated by MC^2-3. Shift code can generate multigroup cross section libraries in ISOXML formats directly (Pandya et al., 2022). They are used to evaluate macroscopic cross sections with interpolation based on variables from other physics and mixing with isotope densities that may or may not depend on other physics. In this regard, ISOXML is initially developed to support the two-step reactor analysis method, where cross section generation and full-core transport calculation are separated into two steps. The format allows multiple data sets in a single XML file that can be assigned to different regions of a reactor and allows for the flexible indexing of individual types of cross sections. It is noted that, although the data sets could be specifically

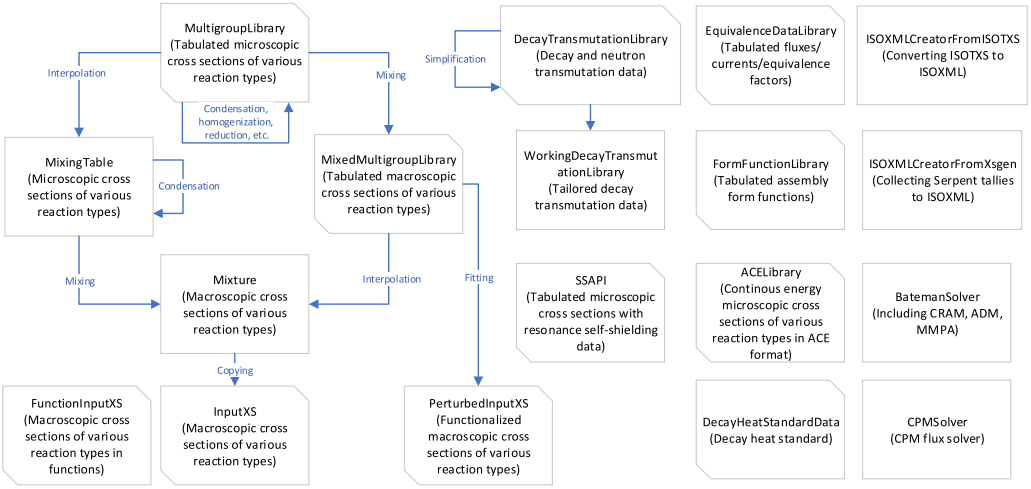


Fig. 5. ISOXML classes.

for a reactor model, the data sets themselves in ISOXML contain no information about reactor geometry.

We adapted an object-oriented programming paradigm for ISOXML with C++ (i.e., data are stored in C++ classes and operations are done through class member functions and construction of classes). ISOXML classes are presented in Fig. 5. Short descriptions of these classes are included in the figure. Classes associated with an XML data format are indicated by the shape with a snip diagonal corner rectangle. Connectors indicate the construction route or operation. For example, the "MultigroupLibrary" class can perform condensation, homogenization, or reduction to generate a new "MultigroupLibrary". A "MixingTable" that contains multigroup microscopic cross sections of all isotopes can be condensed with neutron scalar fluxes into a new "MixingTable" with one-group reaction rates for depletion calculations. A similar design as the multigroup cross section library was later extended to equivalence data, which contains the integrated fluxes for regions or integrated partial currents interfacing regions and boundaries as the inputs for the super-homogenization (SPH) factor or Discontinuity Factor (DF) calculations. In addition, it stores the Griffin computed SPH factors and DFs for a particular mesh and library. On the ISOXML side, the homogenization equivalence data are managed with IDs like the multigroup cross sections. It is up to the radiation transport module to assign the data to mesh regions. The homogenization equivalence regions are typically not the same as the material regions and have finer spatial resolution. We also added a decay and transmutation data format and the Bateman solver in ISOXML to support depletion calculations. It is noted that the transmutation data only contains the transmutation paths with various neutron reaction types but not the corresponding cross sections, which should be in a separate microscopic multigroup cross-section library. We extended the multigroup data format to support different particle types, such as photon, phonon, thermal radiation, etc. For a coupled neutron-photon calculation, two libraries, one for neutron and another for photon, containing the photon production matrix and neutron production matrix, are needed. We also let ISOXML manage empirical decay heat standards (Anon, 2005, 2019, 2014).

3.2.2. SSAPI integration

Methods for multigroup cross-section generation for thermal reactor analyses have been developed in various forms (Duderstadt and Hamilton, 1976). In traditional two-step methods, assembly-homogenized cross sections are pregenerated as a function of core state parameters (temperature, burnup, boron density for LWRs, etc.) before full-core neutron transport (mostly diffusion) calculations. This approach requires homogenization equivalence (Kavenoky, 1978; Smith, 1986) to preserve reaction rates. Assembly Discontinuity Factors (ADF) can be calculated based on a single assembly calculation, a color set calculation, or even a full-core calculation. Alternatively, a SPH factor approach can be used. Both require more calculations than the cross-section generation, which poses a burden to users and brings additional verification points of such equivalence parameters for their validity on various scenarios of transient problems.

As an alternative to this two-step approach, a one-step approach gained attention thanks to increasing computing power (Cho et al., 2007; Kochunas et al., 2017; Ryu et al., 2015; Choi et al., 2015). Multigroup cross sections are generated on the fly for every local region via the self-shielding calculation, and the whole core transport calculation is performed in the same heterogeneous mesh. This direct whole core calculation approach ensures a high-fidelity pin-resolved solution and does not require a homogenization equivalence rather only a condensation equivalence. This condensation equivalence only depends on local pin quantities (Park and Joo, 2017), and thus the condensation equivalence data can be generated easily in the self-shielding calculation (Jae and Joo, 2023). Another advantage of this approach lies in its better capability to capture local multiphysics effects (Doppler, density, etc.) during a transient as the cross sections are updated on the fly at the exact core state every time such effects are in place. Despite these advantages, this approach is not yet suitable for routine design process mainly due to its large computational burden. However, it can be valuable to produce reference solutions for improving models in simpler and faster approaches.

SSAPI was integrated into ISOXML to support either the lattice physics calculation in the two-step approach or the direct whole core calculation in the one-step approach. It was later extended with a double heterogeneity treatment and with methods for the streamline-based PBR analysis. SSAPI requires two data sets: the base multigroup library that contains isotope-wise microscopic cross sections as a function of temperature and background cross section and point-wise cross sections for the self-shielding calculation. Griffin will distribute a universal base multigroup library for any thermal reactors in a fine group structure (~400 groups), and an efficient on-the-fly condensation scheme will be implemented for practical computation times. Alternatively, users can generate their own base multigroup library in a more optimized group structure (~50 groups) tailored for a particular reactor design. NJOY (Macfarlane et al., 2017) was introduced as a git submodule for this purpose, and a set of PyNJOY (École Polytechnique de Montréal, 2015) scripts for specific Evaluated Nuclear Data File (ENDF)/B-based releases automate the procedure for preparing the fine group libraries.

SSAPI provides two options for the self-shielding calculation (Park et al., 2021): the heterogeneous effective cross-section table approach based on the TONE's algorithm (Lee and Jung, 2018a) and the on-the-fly slowing down approach based on the Equivalence Dancoff-factor

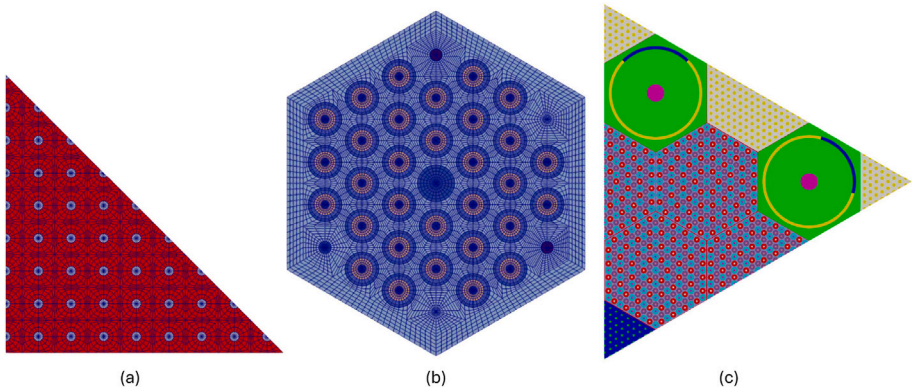


Fig. 6. (a) C5G7, (b) HTTR assembly, (c) 1/6 EMPIRE core.

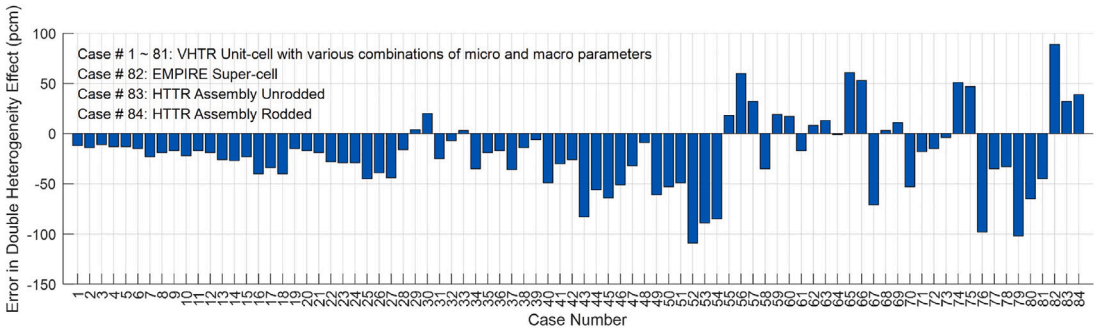


Fig. 7. Errors in double heterogeneity effect predictions over a variety of problems.

Cell (EDC) method (Sugimura and Yamamoto, 2007). The first approach requires several DFEM-SN fixed source calculations to calculate background cross sections, and effective cross sections are calculated via table interpolation. This approach is applicable to any geometry, so it is suitable for Griffin based on the unstructured mesh, as shown in Fig. 6. The second approach performs a slowing-down calculation on the fly in a simplified (macro) unit-cell geometry. The double-heterogeneity option for TRISO fuel is available only in the second option. It involves an additional slowing-down calculation in a TRISO particle (micro) unit cell for flux-volume-weighted homogenization prior to the slowing-down calculation in the macro unit cell. The TRISO unit-cell slowing-down problem is solved using the Iterative Local Spatial Self-shielding (ILSS) method (Park et al., 2022b), which turns out to be very effective in addressing the stochastic particle shadowing effect on spatial self-shielding in a TRISO particle of interest. This is supported by errors of k-effective in the 100 pcm range over a variety of problems, compared to Monte Carlo solutions, as shown in Fig. 7.

3.2.3. Decay Transmutation Library (DTL) simplification and improvement on Bateman solvers

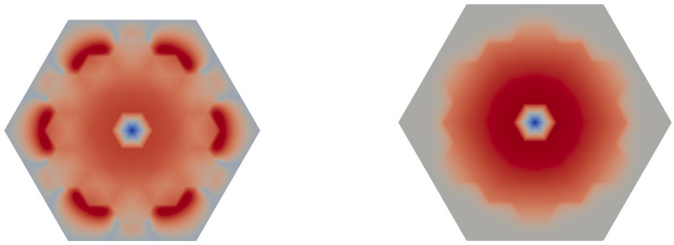
ISOXML allows users to load a DTL with an arbitrary number of isotopes. A DTL contains all quantities in the Bateman equation, including decay constants, branching ratios of all decay types, neutron-induced transmutation pathways, fission yields, etc. Microscopic multigroup cross sections of the neutron reactions of isotopes are included in separate cross-section libraries. The energy distribution of secondary particles from decay is included in a DTL for supporting multi-radiation transport and decay heat calculations.

At its extreme, DTL can contain 3,820 nuclides from the ENDF/B-VIII.0 library. However, to maximize performance without degrading accuracy, less important isotopes can often be removed from the library. For typical reactor depletion calculations, several hundreds of isotopes are sufficient. For example, DRAGON uses a library with 300 isotopes. ISOXML can simplify the DTL from the ENDF/B-VIII.0 data set with a heuristic method to meet certain criteria, including accurately preserving decay energy (Calvin, 2023). Adding pseudo-nuclides for decay energy tracking can significantly reduce the number of nuclides needed by the simplified depletion library to roughly 350 nuclides with 10 pseudo-nuclides.

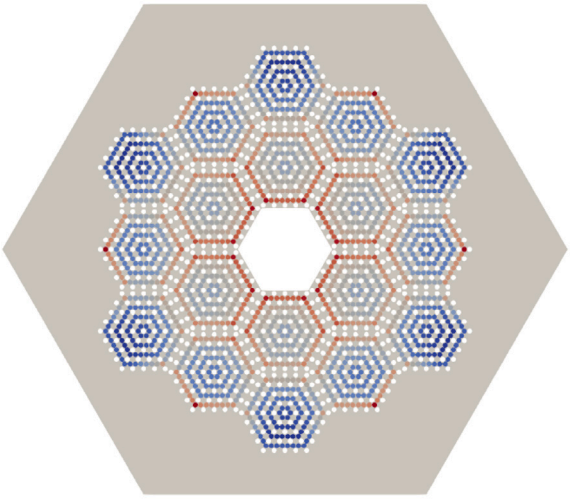
ISOXML also introduces numerical isotopes (Calvin et al., 2023a) whose time derivatives depend on atomic densities, such as time-integrated fission rates, damage energy, decay energy, etc. These numerical isotopes are solved along with normal isotopes in the Bateman solver, such as Chebyshev Rational Approximation Method (CRAM), which provides better time integration. A new Bateman solver, the Adding and Doubling Method (ADM), was introduced in ISOXML (Calvin et al., 2023b). The Gauss-Seidel (GS) method for solving the CRAM system of equations was introduced to improve performance compared with the sparse Gaussian elimination (SGE) method for practical purposes (Calvin and Choi, 2023).

3.2.4. Assembly form function data

In order to support homogeneous calculations requiring pin power reconstruction, form functions are an important step for recovering pin-level heterogeneous reaction rates that are otherwise lost through the assembly homogenization process (Kumar et al., 2024, 2022; Prince et al., 2022). In order to support such calculations and assist with the data management relevant for storing pinwise and groupwise form functions, an internal library for storing form function data has been created in ISOXML. Helper functions in ISOXML have been defined to automatically normalize pin reaction rates for each pin and energy group from a single assembly, colorset, or full-core heterogeneous calculation and populate the form function data library. Single assembly form function data assume flat homogeneous assembly core conditions with reflective boundary conditions at the assembly boundary, while form functions calculated from colorset or full-core calculations capture flux gradients in the homogeneous system by running a separate Griffin calculation to compute pinwise and groupwise pin reaction rates for the



(a) Homogeneous thermal flux (b) Homogeneous fast group flux



(c) Reconstructed pin powers in fuel pins

Fig. 8. Illustration of the homogeneous flux distribution and reconstructed pin powers for the Empire core.

homogeneous system. This Griffin calculation can be combined with the equivalence factor generation step, and the form function library data for each pin and the energy group will be updated accordingly to account for the flux changes in the homogeneous system. Form function data can also be functionalized in terms of input state variables.

In addition to the form function data, the form function library representation in ISOXML also stores parameters related to the pin geometry (radius and pitch) as well as the structured pin lattice layout (orientation and number of rings for hexagonal pin lattices and number of columns and rows for Cartesian pin lattices). By storing this information as a separate file, the input structure of the pin power reconstruction module in Griffin is greatly simplified, and the user simply needs to map the "assembly ID" extra element integer representing each assembly region on the homogenized Griffin mesh to a form function library ID defined by ISOXML to represent the form functions and pin lattice layout for each homogenized assembly region. Fig. 8 highlights the thermal and fast group flux distribution of the homogeneous system for the Empire core, as well as the heterogeneous reconstructed pin powers using assembly-based form function data. More detailed information about the form function data library layout and how form functions are defined can be found in Ref. Park et al. (2022a).

3.3. Radiation transport

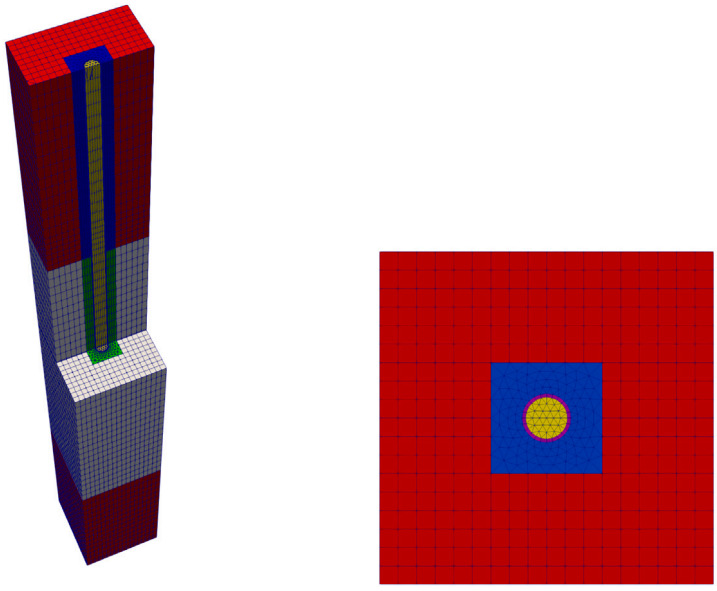
3.3.1. General features

The radiation transport module supports both the strongly- and fully-coupled multiphysics calculations, as described in Section 3.1 and explained in Ref. Wang et al. (2021b). It supports transport calculations for various forms of radiation, including neutron, photon, thermal radiation, phonon, electron, or multi-radiation transport. The transport equations are split into terms in their weak forms referred to as kernels, boundary conditions, discontinuous Galerkin kernels, interface kernels, mortar constraints, etc. Some of these terms are shared by all forms of radiation, such as streaming, collision, scattering, vacuum/reflecting/white boundary conditions, etc. Particular kernels can be added, such as fission kernels for neutron transport or black body emission for thermal radiation transport. This design makes supporting any new form of radiation straightforward. Coupled radiation transport calculations, such as coupled neutron-photon calculations, are also supported.

The radiation transport module utilizes various FEM-based spatial discretization modalities: Continuous Finite Element Methods (CFEM), DFEM, and HFEM. The angular discretization can be treated with the diffusion approximation, PN, and SN. Griffin provides a point-kinetics equation solver based on the MOOSE scalar variable system. The primary solvers supported and their applications are shown in Table 1. Two HFEM schemes are new and will be discussed in a separate section. Each of these schemes offers some unique features and can be attractive for certain applications. SN schemes are suitable for high fidelity heterogeneous calculations while PN or diffusion schemes are suitable for calculations with spatial homogenization. Compared with diffusion schemes, PN schemes offer a treatment of pronounced streaming, the effect often encountered in fast reactor analyses. In PN schemes the computing effort increases almost quadratically with the number of angular moments, thus PN schemes are typically used with low PN order, rarely exceeding five although the implementation in Griffin does not have a PN order limitation. All schemes can handle arbitrary numbers of groups and scattering orders and all support eigenvalue, transient, depletion, and adjoint calculations with parallelization through domain decomposition. All schemes can perform evaluations with mesh displacement. Small functionality gaps in the schemes might exist but

Table 1
List of Griffin transport schemes and usage (Wang et al., 2021b).

Solver	Usage	Typical number of elements	Curvilinear coordinate
CFEM-Diffusion	Assembly/pincell homogenized	$< 1M$	Y
HFEM-Diffusion	Assembly/pincell homogenized	$< 100k$	Y
HFEM-PN	Assembly/pincell homogenized	$< 100k$	N
DFEM-SN	Heterogeneous	$100k - 100M$	Y



(a) Cut view of the mesh. (b) X-Y view of the mesh.

Fig. 9. Mesh of 43,560 elements used for the 3D TREAT benchmark problem with multi-scheme calculations. The gray region represents the active core region. Reflecting boundaries are applied on all X and Y boundaries. Vacuum boundaries are applied on the top and bottom boundaries.

can be filled quickly due to the modular design. DFEM- and HFEM-based methods allow solution discontinuity across element interfaces and offer advantages for reactor analysis, often with material heterogeneity. Nonlinear Diffusion Acceleration (NDA) (Wang et al., 2014; Schunert et al., 2017) and Linear Diffusion Acceleration (LDA) (Prince et al., 2020; Wang et al., 2022) are available for SN schemes and provide improved performance for heterogeneous reactor analyses. A coarse-mesh NDA or LDA with finite difference, often referred to as a Coarse Mesh Finite Difference (CMFD), is available for the DFEM-SN and HFEM-PN methods. The Improved Quasi-Static (IQS) method is an temporal acceleration scheme for transient calculations. All these schemes can be involved in the so-called multi-scheme calculations, where the solution domain is covered by multiple subregions with various levels of spatial mesh resolution and each subregion utilizes its best-performing transport scheme to maximize computing performance (Wang et al., 2017). An example of a multi-scheme calculation is illustrated in Fig. 9 of a 3D Transient REActor Test (TREAT) mockup model in Ref. Labouré et al. (2022b), where the center assembly with a control rod is meshed in detail and covered with the SN scheme while the other eight assemblies are homogenized with diffusion. Two types of interface conditions, mortar and penalty approaches, are available in multi-scheme calculations.

The radiation transport module includes the transport materials that interact with ISOXML to evaluate cross sections. Feedback through thermal expansion, density, and temperature changes are inherently handled in all Griffin materials regardless of the nuclear data type. On-the-fly group condensation (Saller et al., 2017), control-rod or control-drum cusping treatment (Schunert et al., 2019), local mesh volume adjustment, etc. are also available regardless of the nuclear data type. It is noted that mesh displacement, density, and temperature variables are from solid mechanics, fluid dynamics, etc., which can be modeled with MOOSE-based or non-MOOSE-based applications and coupled though MultiApp/Transfer for fully-coupled multiphysics simulations.

Transport schemes can accept homogenization equivalence factors based on either SPH or DF or a combination of both. Griffin deploys a PJFNK solver to resolve the nonlinear dependency of the flux and equivalence factors during the calculation of the equivalence factors, which typically outperforms traditional iteration techniques (Ortensi et al., 2018; Labouré et al., 2019). Homogenization equivalence has been extended to multi-scheme calculations (Labouré et al., 2022b). Although homogenization equivalence is a typical reactor analysis step of preparing cross sections, we put homogenization equivalence in radiation transport because it is closely integrated into transport schemes. Delayed Neutron Precursor (DNP) convection was added to support fluid-fueled systems and is available to all neutron discretization schemes. The implementation allows users to model DNP outside of the radiation transport module along with fluids, thus the radiation transport module can either take the velocity or direct DNP concentrations from the fluid (Jaradat et al., 2024). The Pebble Tracking Transport (PTT) (Wang et al., 2018a) was added with DFEM-SN and was recently improved via CMFD acceleration (Wang et al., 2024) for high resolution PBR analyses. With PTT the pebble-bed heterogeneity, pebbles and gap, is explicitly resolved without the need to mesh the surface of each pebble individually. Nodes in the pebble-bed region correspond to pebble centroids, and each tetrahedron has five pieces: four pebbles at the four vertices and the gap in between, illustrated in Fig. 10 from Carlson et al. (2024) for a generic 200 MWt PBR model.

3.3.2. Heterogeneous calculations with DFEM -SN

Mesh sweeping is the base-stone for solving the DFEM-SN system. It is a matrix-free solving method for a convection–reaction equation

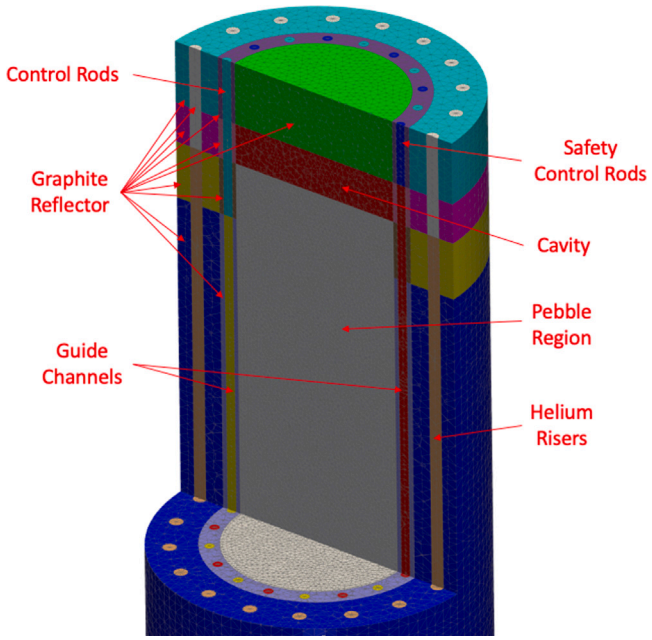


Fig. 10. Mesh of a generic 200 MW PBR for PTT.

discretized with the upwinding technique, including DFEM-SN in Griffin. Mesh sweeping has been investigated significantly in the past, for which we list a few references here: (Mathis et al., 2000; Pautz, 2002; Plimpton et al., 2005; Vermaak et al., 2021; Adams et al., 2020). Performance is often the main concern in these studies. However, our main consideration in Griffin is to have a sweeper residing in a multiphysics environment to support any given domain decomposition without lagging information on processor interfaces during sweeping. The domain decomposition is dictated by the overall simulation, including all participating physics, thus tuning it with techniques such as Koch-Baker-Alcouffe (KBA) (Kock et al., 1992) is not applicable and not the focus. A sample domain decomposition is showed in Fig. 11 with the 2D Empire microreactor mesh, where the entire domain with 402,444 elements is partitioned with METIS (George and Kumar, 1997) into a total of 96 partitions. We used the MOOSE ray-tracing module for task scheduling and directly developed it with libMesh for performance. We used the simple First-In-First-Out (FIFO) task scheduling technique and use buffers to reduce the number of communications for better parallel performance. Readers are referred to Ref. Prince et al. (2024) for more details. Currently, we make the sweeper accept a source with the same number of degrees of freedom as in the representation of the angular flux, which is the primary variable of DFEM-SN, so that it can directly take the residual or source vector passed by a preconditioner. Element-averaged total cross sections, which depend on local material compositions and may depend on multiphysics variables, are evaluated as the by-product of the residual evaluation and are used during sweeping. The sweeper can operate on an unstructured mesh with any types and shapes of elements supported by libMesh. We use the averaged normals for curved element sides to determine the upwinding–downwinding relation of two neighboring elements. Elemental mass matrix and convection matrices are preassembled and used during the sweeping. These elemental matrices are energy-group and streaming-direction independent, thus their storage is not an issue in terms of memory usage. Using the 2D Empire as an example, the grind time obtained with the INL Sawtooth machine for mesh sweeping, defined as the total CPU time in mesh sweeping (49.3 s) divided by the number of mesh sweeps (60) and the number of total unknowns (637,471,296) multiplied by the number of processors (96) is about 0.12 microseconds while the grind time for 12 residual evaluations is roughly 0.2 microseconds on the unstructured mesh in Fig. 4 and with the domain partitioning in Fig. 11. Grind time for 3D calculations is larger than for 2D calculations, but the numbers given here are representative. Parallel efficiency of the sweeper with 1536 processors for the Empire 3D benchmark that has 6,841,548 elements, 11 energy groups and 48 streaming directions, is about 80% (Wang et al., 2021a). Parallel efficiency with 1536 processors for the Empire 2D benchmark with 402,444 elements, 11 groups and 48 streaming directions can still reach 45%. The sweeper also supports R-Z curvilinear coordinates.

To further boost the performance and improve the user interface, we reimplemented the accelerated Richardson iteration. Motivated by the MOOSE PJFNK solver, we adapted a residual-based Richardson iteration, illustrated in Fig. 13, where one iteration is done by a residual evaluation followed by a transport update. Residual evaluation is done through kernels that can bring in many types of feedback, such as thermal expansion or Doppler with fuel temperature. The transport update takes the residual as the input and returns an updated solution closer to the converged solution. Typically, we can perform one sweep to get the product of the inverse of the streaming-collision operator and the residual as the correction. It is noted that this transport update routine can be used as a preconditioning technique for the PJFNK solver. To have faster convergence, an acceleration step with either NDA or LDA within a Richardson iteration needs to be included. We implemented a meshless CMFD, where the CMFD system is assembled algebraically and solved with PETSc. The CMFD is meshless because it only needs an extra element ID for indicating the coarse element ID of elements on the mesh, rather than requiring a valid coarse mesh. This design allows users to use a regular grid to assign coarse element IDs for CMFD, i.e. elements whose centroids fall into the same grid have the same coarse element ID, thus having a coarse mesh is not required; although, users are allowed to use a coarse mesh to assign coarse element IDs. A sample coarse element ID assignment with 1370 coarse elements for the 2D Empire microreactor mesh is illustrated in Fig. 11 by overlaying a regular Cartesian grid. We can see irregular coarse element interfaces if we zoom into the figure. This coarse grid delivers similar convergence behavior as an independently developed coarse mesh with 2160 coarse elements. The CMFD system allows users to use a smaller number of CPUs to reduce the communication cost if the system is small. For example, the CMFD system is solved with eight processors for the 2D Empire microreactor model determined by Griffin automatically. CMFD was also made to support curvilinear coordinates, mesh displacement, etc. It is noted that the CMFD implementation is flexible enough that HFEM-PN can also use it. When using CMFD, the typical sweep update strategy may not work effectively, thus we added a matrix-free multigroup iterative solver to bring scattering into consideration for the transport update. This multigroup iterative solver is based on angular flux moments, and the group sweep is arranged in stages, where all energy groups of the same stage can be updated simultaneously. Traditional source iteration or Generalized Minimal Residual method (GMRes) solving within each stage and thermal iteration for stages containing upscattering are implemented. Because the transport update does not need to be exact, the truncation of the scattering term is allowed and almost does not affect the overall Richardson iteration convergence but with a faster update operation. This solve technique is extended to the transient with the IQS and to depletion with the criticality search (Prince et al., 2024). It is also extended for PTT (Wang et al., 2024). This accelerated Richardson iteration is implemented with the latest solver system available in the MOOSE framework, illustrated in Fig. 13, which contains three more solve objects for criticality search, multiphysics and depletion. A latest steady-state eigenvalue calculation with the accelerated Richardson iteration for a 3D Advanced Test Reactor (ATR) model with 7,393,350 elements, 18 energy groups and 64 streaming directions, that end up with total ~34.06 billions of degrees of freedom, took about 100.7 min on the INL's Sawtooth machine with 6 nodes and with 24 processes per node (Nguyen et al., 2024).

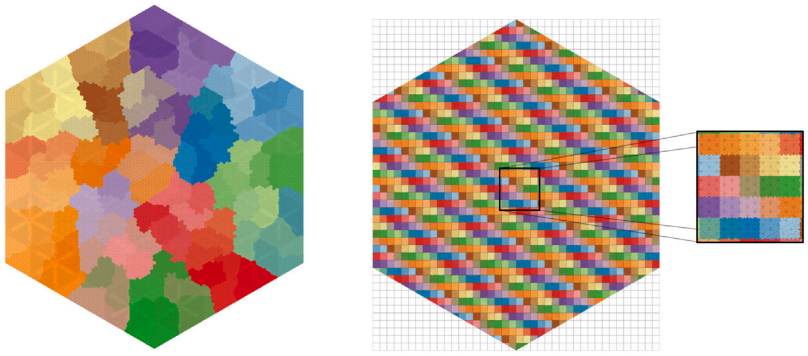


Fig. 11. Illustrations of the mesh partitioning and meshless CMFD capability in Griffin for the 2D Empire microreactor mesh.

3.3.3. Homogeneous calculations with HFEM-PN

The PN scheme allows the treatment of pronounced streaming important for fast reactor analyses. As mentioned earlier, PN is not recommended for use with very high PN orders because the computing cost increases quadratically with the number of angular moments. The PN angular convergence could be slow if the solution is not smooth, which is often the case for problems with heterogeneous configurations, but PN is preferred for problems with significant spatial homogenization with fewer unknowns in the angular discretization. In fast reactors, angular flux distributions within assemblies tend to be relatively smooth, owing to the long mean free path of fast neutrons. Anisotropic scattering introduces a notable transport effect throughout the core. These distinctive characteristics have led to the wide adoption of PN in fast reactor analyses.

Among various FEMs, HFEM offers p-refinement by increasing the number of expansion polynomials on elements and a fast red-black iteration algorithm that takes only partial currents on element interfaces. Its application has been demonstrated in the past, known as Variational Nodal Method (VNM) (Diber and Lewis, 1985; Yang et al., 2001; Smith et al., 2003; Wang et al., 2011; Li et al., 2018; Zhang and Li, 2022). The default PJFNK solver for other CFEM-based PN schemes performs poorly due to the lack of good preconditioning. Thus, we decided to implement HFEM-PN in Griffin. We also added HFEM-Diffusion along the development as an alternative solver for existing CFEM and DFEM diffusion solvers because it shares the same infrastructure as HFEM-PN in the MOOSE framework and required minimal effort for implementation. The dedicated diffusion solver option is more efficient because it can skip spherical harmonics related operations required in PN.

The implementation in Griffin follows the same paradigm as other schemes. We have the full residual evaluation that can take care of the multiphysics coupling. We have the red-black iteration for updating and diffusion acceleration. This approach eliminates the need to evaluate response matrices whenever cross sections are changed due to depletion or the change in multiphysics environment variables. Instead, response matrices can be assembled at the beginning of a simulation with the reference cross sections. In essence, the red-black iteration plays the same role to HFEM-PN as the mesh sweeping to DFEM-SN. Red-black iteration is also used in the PJFNK solver as a preconditioner. However, diffusion-accelerated Richardson iteration with red-black iteration typically outperforms the PJFNK solver with a red-black preconditioner. To further improve efficiency, we introduced the GMRes method to the red-black iterations in a matrix-free manner (Lewis et al., 2013). The anisotropic scattering contributions are explicitly treated with PN, which has a significant impact on accuracy in SFR analysis. The HFEM-PN solver can be applicable to conventional lattice configurations as well as an unconventional configurations with unstructured meshes. Fig. 12 demonstrates the unstructured mesh capability by performing the HFEM-PN calculation for a hexagonal lattice with arbitrary radial deformations requiring unstructured elements. To provide the complete capabilities for reactor analysis, we implemented the pin-power reconstruction scheme, which enables retrieving pinwise quantities from a spatially-homogenized solution with a pregenerated form function set (Smith, 1986). The diffusion option supports the calculations of homogenization equivalence with SPH and DFs.

3.4. Reactor analysis

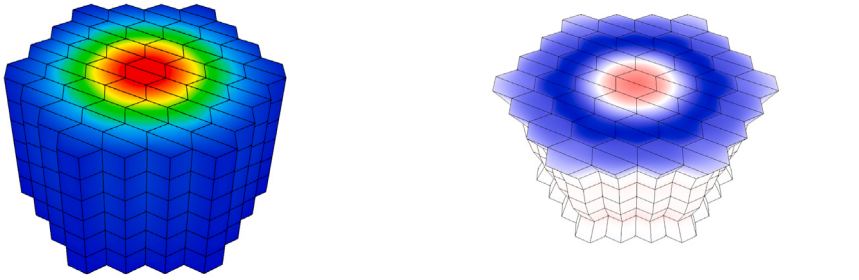
3.4.1. Depletion capability

Depletion is an essential capability required to model power reactors where the build-up of actinides and fission products can significantly affect the neutronic behavior. It is a unique calculation in neutronics in the sense that it is a transient calculation but the transient is so slow that the fluxes are obtained through steady-state eigenvalue calculations at each time step and correspondingly the total power is an input for the flux normalization. Only the atomic densities, burnup, etc. have the time derivative term in their equations. Griffin supports depletion calculations with some unique features.

Griffin allows a flexible spatial resolution by using an extra element integer on an unstructured mesh. All elements with the same element integer are considered together as a depletion zone. Using LWRs as an example, a depletion zone can be an assembly, a fuel pin, a ring of a fuel pin, or even finer. One important constrain is that the compositions of all elements in a depletion zone must be the same.

Griffin also allows both macroscopic and microscopic depletion. In macroscopic depletion calculations, macroscopic cross sections are computed and tabulated in a grid of burnup values among other grids. The preparation of suitable cross sections can be performed by external lattice codes, such as DRAGON (Marleau et al., 2011) or external Monte Carlo codes, such as Serpent (Leppänen, 2015; Pandya et al., 2016; Romano et al., 2015). Burnups are accumulated across the individual depletion zones. Macroscopic cross sections are linearly interpolated using the local burnup value. In contrast to macroscopic depletion, microscopic depletion computes the atomic densities of the isotopes in the materials of interest from solving the Bateman equation. The macroscopic cross sections required by radiation transport calculations are then computed using the atomic densities and microscopic cross sections. A DTL, containing isotope decay data and transmutation with neutron reactions, is required for microscopic depletion. Griffin allows the cross-section evaluation to be separate from the isotope tracking (i.e., cross sections can be evaluated as in a macroscopic depletion while isotopes are tracked with a simplified microscopic cross-section library and DTL). We plan to incorporate the online self-shielding treatment with microscopic depletion in the future.

Giffin also allows decay heat to be tracked during depletion. The decay heat can be either evaluated explicitly with all the tracked isotopes when microscopic depletion is turned on or can be evaluated with empirical decay heat models. For each depletion zone, necessary



(a) Original mesh with flux solution (b) Deformed mesh with flux changes caused by deformation

Fig. 12. Illustration of an unstructured mesh created through mesh deformation.

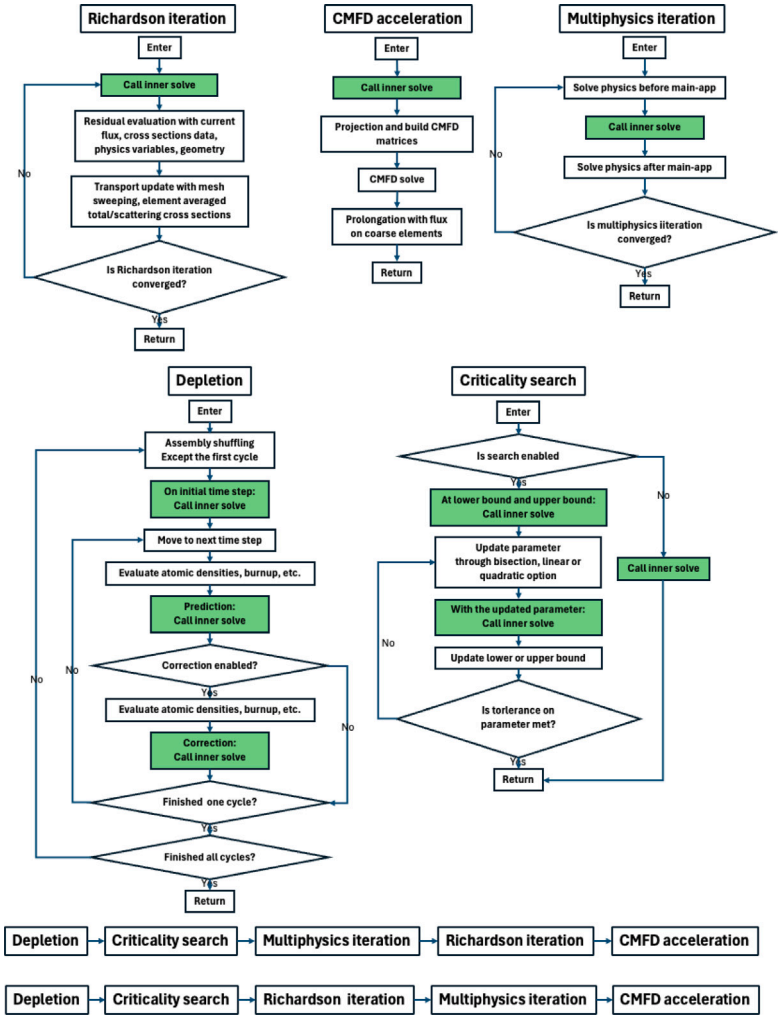


Fig. 13. Illustration of the depletion algorithm in Griffin.

quantities, such as burnup or operation time, for the decay heat models are evaluated. When decay heat is tracked, the total power is split into the prompt power, which is evaluated with the neutron flux and energy deposition cross sections, including the contributions from fission and non-fission reactions, and the decay power. The power at a particular time, specified as a power modulation function on a rated power, is subtracted from the decay power and is then used to normalize the flux obtained from the eigenvalue calculation. The rated power cannot be smaller than the decay power, but for convenience, Griffin allows this. In such a case, the prompt power scaling factor will be set to zero. When decay heat is not tracked, fluxes are normalized based on the total recoverable fission power with the underline equilibrium assumption.

The depletion calculation is one of the most complicated calculations in Griffin partly because of the cyclic dependency of the evaluation sequence at a time step. With macroscopic depletion, the accumulation of burnup (integrated power) of a depletion zone is evaluated with the local power, which depends on the fission cross section and energy emitted per fission and the flux. Cross sections, including the fission cross sections, are tabulated with burnup and affect the flux distribution. With microscopic depletion, the atomic densities are obtained by solving the Bateman equation, which depends on and affects the

fluxes. The power scaling factor introduces another cyclic dependency as well. We resolve this cyclic dependency with a classic predictor–corrector scheme. The power scaling factor, atomic densities, burnup, and all necessary quantities are evaluated in a single function and can be called at the beginning of the prediction and correction steps, which takes the current fluxes, the quantities of the previous time step, as the input. The same predictor–corrector algorithm can drive both microscopic and macroscopic depletion with and without decay heat. The full algorithm is illustrated in Fig. 13. There are five solve objects, as showed in the graph: depletion, criticality search, Richardson iteration, CMFD acceleration, and multiphysics iteration. A solve object may have zero or one inner solve object. The Griffin depletion solver currently allows two arrangements of these five solve objects. The depletion solve object is always the outermost object, followed by the criticality search, which can be optionally turned on and off. CMFD acceleration is the innermost object. The Richardson iteration and multiphysics iteration ordering can be swapped. An outer multiphysics iteration is preferred when the Richardson iteration for neutronics is relatively less costly than other physics, while an inner multiphysics iteration is better when the neutronics Richardson iteration is heavy. The green boxes in Fig. 13 are the places where the immediate inner solve object of individual solve objects are called. For diffusion calculations, the Richardson iteration accelerated with CMFD can often be replaced with the MOOSE PJFNK solver calling PETSc, which is implemented in a solve object residing in MOOSE. All other solve objects for depletion calculations can be reused. The multiphysics iteration solve object is used in typical steady-state and transient calculations and is part of MOOSE. The multiphysics iteration solve object is for fully-coupled multiphysics calculations. When doing strongly-coupled multiphysics calculations, other physics are solved together with neutronics and the solve object does not have the multiphysics iteration and will simply pass the call to its inner solve object.

As a final note, it is important to say that PBRs and MSRs have different depletion characteristics and need to be handled differently since the fuel moves during operation in both reactor types. Depletion modeling for PBRs is discussed in Section 3.4.4. PTT will be used to support high-resolution running-in calculations tracking the atomic densities of individual pebbles in the future. We will need to differentiate the short-lived isotopes from the rest of the MSR inventory because their distribution can be affected by the flow. Isotopes can be considered in a single Bateman solve with the integrated reaction rates accounting for the density distribution in the core solve. The transmutation of short-lived isotopes while they are flowing through the core in seconds can possibly be neglected, as their densities are likely saturated at a depletion step lasting days. Decay heat precursors can be short-lived, and their convection is important for accurately obtaining the decay heat distribution within the core. The convection of short-lived isotopes, including delayed neutron precursors and decay heat precursors can be optionally modeled on the fluid side along with other effects such as deposition, multiple phases, etc. Griffin is able to obtain distributions of those short-lived isotopes in the core if they are modeled externally. Online isotope extraction and fuel insertion can be modeled in the Bateman equation as a sink and source. The depletion calculation can be achieved with the same predictor–corrector algorithm.

3.4.2. Fuel management

This capability is developed to support reactor designs that have multiple-cycle operations with assembly shuffling between two successive cycles. This capability is required to model reactors with fixed solid fuel arranged in assemblies, such as LWRs and various fast reactor designs. This is not for PBRs or MSRs because their fuel management is different. Griffin has a flexible assembly-based fuel management scheme. This fuel management scheme supports both heterogeneous and homogeneous reactor models. To perform fuel management, we utilize an element integer as the assembly ID on an unstructured mesh in Griffin. Elements with the same assembly ID are contained as one assembly and are considered together during fuel management. The assembly IDs of all mesh elements are typically assigned during the mesh generation procedure.

Griffin checks a few constraints:

- A depletion zone must not cross assemblies (i.e., a depletion zone must be contained within one assembly);
- The mesh for the current assembly and the assembly to be shuffled must have the identical arrangement of mesh elements; in other words, fuel shuffling must not change the mesh;
- A one-to-one mapping of depletion zones must be satisfied in the current assembly and the assembly to be shuffled.

and will issue errors if these constraints are not met.

If the shuffling assembly is a fresh assembly, atomic densities are initialized based on the compositions of the assembly type. If the current assembly is discharged, the depletion data of that assembly is saved in a binary file, that can be loaded for later consideration. Shuffling does not change the assembly and depletion IDs. Griffin provides a 2D map to help users specify the fuel shuffling pattern, as illustrated in Fig. 14. The index in the 2D map indicates the assembly location, and the number of the 2D map is the ID of the assembly to be moved into this location. If a dash sign is provided, it means that this assembly is not moved during shuffling. If the number is negative, it means a fresh assembly is to be used, and the absolute value of the number is the assembly type that users added in the mesh generators. Each cycle can have an associated assembly shuffling pattern, so Griffin knows at the end of the cycle which pattern is to be invoked. We remind users that, after each assembly shuffling, binary files for storing the depletion data of discharged assemblies are generated. Griffin supports loading the depletion data in binary files and performing separate decay calculations for the loaded depletion data.

3.4.3. Two-step method for SFR analysis by integrating MC^2-3

ANL has established a series of modeling capabilities for fast reactor analyses with MC^2-3. We found that the conventional workflow is quite tedious and error-prone in that preparing MC^2-3 inputs is difficult, especially when populating atomic densities for all inputs and creating the whole-core-equivalent RZ geometry for treating the streaming effects caused by the long mean-free-path of fast neutrons. There could be a potential mismatch between the geometry used for MC^2-3 calculations and the full-core Griffin calculations. To resolve this issue, Griffin provides an optional simplified workflow by linking to MC^2-3. Griffin uses the RGMB provided by the MOOSE reactor module (Shemon et al., 2023) to describe the geometry and generate a mesh for MC^2-3 calculations as well as full-core calculations. In Griffin, material compositions, each consisting of a set of isotopes with their isotope densities, are defined in the input and passed programmatically to MC^2-3 calculations and full-core calculations. Griffin then uses composition IDs for the composition assignment and solves the RZ ultra-fine group problem with its DFEM-SN solver, which, in previous ANL analysis relied on the external TWODANT (Alcouffe et al., 1990) code. The fast-reactor workflow developed in Griffin allows users to focus on describing the core geometry based on the RGMB mesh and specifying material compositions. The workflow can generate a multigroup cross-section set specific to the user-provided core configuration by employing MC^2-3. Optionally, the core-wide spectrum calculation is performed on the simplified RZ geometry using the DFEM-SN solver, which is for resolving the cross-section region dependency. Then, full-core calculations can be done with HFEM-PN or DFEM-SN solvers, with support of multiphysics, depletion, and fuel cycle capabilities. The overall calculation flow of the cross-section generation workflow is illustrated in Fig. 15.

On top of the existing ANL analysis methods: assembly-homogeneous, duct-heterogeneous, and fully-heterogeneous models, INL spearheaded the development of a new ring-heterogeneous method (Terlizzi et al., 2024) that can model thermo-mechanical feedback

```
[FuelManagement]
  [pattern]
    type = AssemblyShuffling
    pattern =
            '2  5;
          1  -1   7;
            3   6'
  []
[]
```

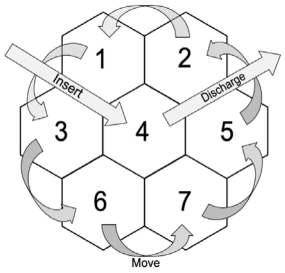


Fig. 14. Example input format for the fuel management scheme.

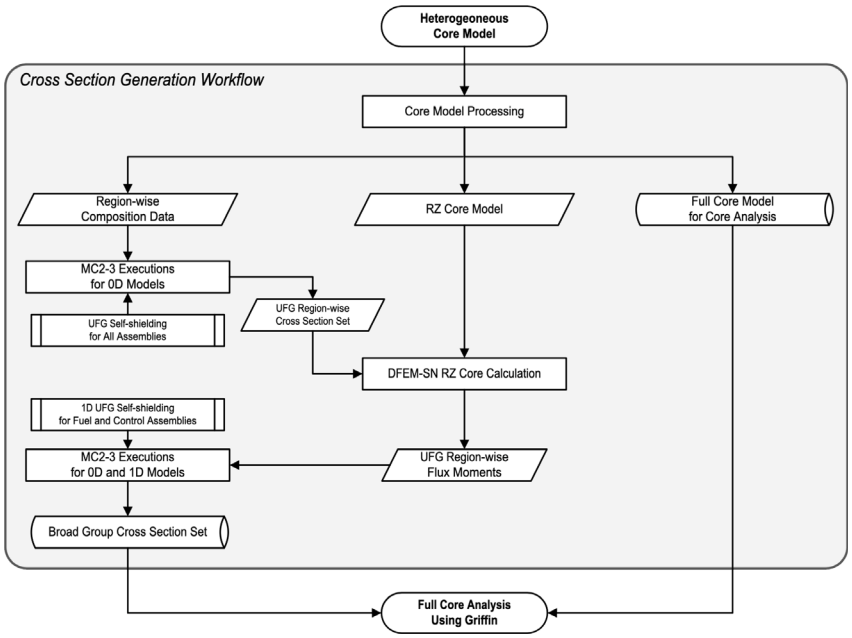


Fig. 15. Calculation flow of SFR cross-section generation.

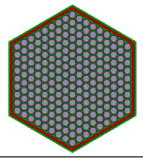
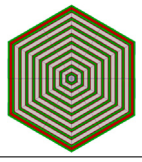
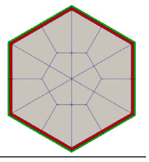
effects more accurately than the homogeneous models. It consumes reasonable computing resources compared with the fully-heterogeneous model, thus making itself suitable for routine design simulations. As with the duct-heterogeneous approach, the ring-heterogeneous approach resides at an intermediate spatial resolution level between the assembly-homogeneous and fully-heterogeneous approaches. However, while the duct-heterogeneous approach homogenizes all components in a given assembly except for the duct and external stagnant sodium regions, the ring-heterogeneous approximation retains the heterogeneity between each component, considering that each reactor component must be able to be explicitly deformed to properly capture the differential material expansion and associated feedback effects in SFRs. The duct-heterogeneous approximation can still explicitly model the radial deformation of assemblies by retaining the heterogeneities in the duct region. However, it is limited in regard to simulating the axial deformation of unit components, such as fuel or control rod drivelines, and it must rely on the cross-section functionalization with respect to the deformation. The ring-heterogeneous approximation is intended to eliminate such limitations.

The idea behind the ring-heterogeneous approximation is to cast an array of pin cells into the "bands" of materials and components constituting the pin cells (e.g., fuel, cladding, and coolant) while still preserving their volumes. In a hexagonal geometry, the pin cells along each hexagon will be smeared into bands that resemble rings, which gives this approximation scheme its name. By doing so, the ring-heterogeneous approximation retains the heterogeneities of all the components yet still significantly reduces the number of elements. Table 2 shows the spatial approximation effect when moving from fully-heterogeneous to ring-heterogeneous and duct-heterogeneous mesh on a single 2D ABTR fuel assembly. The data shows that the ring-heterogeneous approximation can reduce the number of elements from a fully-heterogeneous mesh by 40 times. For a 2D ABTR core problem, the difference in k-effective and ring-wise fission rates between fully-heterogeneous and ring-heterogeneous models is negligibly small, less than 20 pcm and 0.02% in Root Mean Square (RMS) for ring-wise fission rates, respectively (Terlizzi et al., 2024). The generation of a ring-heterogeneous model from a fully-heterogeneous model is fully automated in Griffin along with assembly-homogeneous and duct-heterogeneous models.

3.4.4. Streamline-based PBR depletion

Griffin includes the ability to directly calculate an equilibrium PBR core and the ability to perform "running-in" simulations. The modeling approach used by Griffin for equilibrium core calculations is described in Ref. Schunert et al. (2023) and the extension of this methodology to running-in simulations is described in Ref. Hanophy et al. (2024). Griffin models pebble movement by assuming the movement can be described by a series of 1D streamlines, which has been validated with experiments (Kadak and Bazant, 2004). Pebbles flow entirely within one of the streamlines, entering a streamline at the inlet of the reactor; there is no cross flow between the streamlines. These streamlines are then discretized into segments. A typical Griffin model (used in Ref. Prince et al. (2023)) is shown in Fig. 16. Fig. 16(a) depicts the way streamlines are defined and how they relate to the mesh used for

Table 2
Comparison of fully-heterogeneous, ring-heterogeneous, and duct-heterogeneous mesh on a 2D ABTR fuel assembly.

Type	Fully-heterogeneous	Ring-heterogeneous	Duct-heterogeneous
Mesh			
Number of cells	22,056	546	42

deterministic transport calculations. Streamlines are defined by a user as piecewise continuous lines. These lines pass through the neutronics mesh as shown in the figure, and based on the proximity of mesh cell centroid to a streamline, the mesh cells are partitioned among the streamlines.

A Eulerian modeling approach is used in the resulting control volumes where the conserved quantities are the volume fraction and the nuclide densities of each pebble type. Griffin also explicitly calculates pebble burnup, and so these conserved quantities also depend on burnup. To make the streamline equations fully discrete, the burnup space of interest, from zero to some maximum value, is discretized into so called "burnup bins". The user sets the maximum value for the burnup discretization. The burnup limit beyond which pebbles are discarded instead of being reloaded is usually set below the regulatory limit for the fuel burnup. Since Griffin directly computes burnup, a user can set the maximum burnup past the nominal burnup limit for reloading pebbles and then see the actual burnup distribution of pebbles exiting the reactor. Pebbles "advect" through this burnup space with a "velocity" given by the partial power. The streamline balance equations then look like a 2D advection equation; advection in space along the streamlines and "advection" in burnup space through the burnup bins. Simple upwinding with finite difference is currently used for discretization. The streamline equations are coupled with neutron transport equations. A mapping is used to move data between the mesh used for neutronics calculations and the streamlines. The streamlines accept reaction rates for depletion from the neutronics calculations and provide atomic densities to the neutronics calculation, which uses them to compute macroscopic cross sections. To compute macroscopic cross sections for transport calculations, Griffin can utilize a pregenerated multigroup library as the traditional two-step method or online cross-section generation, which is discussed next.

This pebble streamline approach is supported by the online cross-section generation with SSAPI. Whenever the pebble depletion module updates pebble volume fractions and isotopic number densities per fuel pebble type and burnup group, SSAPI generates spectrum-zone-wise homogenized cross sections in a few groups for the core diffusion or transport calculation and provides pebble-wise (per fuel pebble type and burnup group) isotope-wise one-group reaction rates to the pebble depletion module using the neutron transport solution. This process improves the accuracy of the depletion calculation as the cross sections are generated at improved local conditions. Preliminary verification results showed that the implemented multi-pebble model produces a very accurate solution while the conventional average pebble model widely used in legacy codes inevitably fails to account for different self-shielding effects in different pebbles (Park et al., 2024). At the time of writing, the remaining work is to verify the implementation with a realistic PBR core design, to consider the pebble clustering effect in graphite pebble-mixing cases, and to implement the on-the-fly condensation with a leakage correction technique.

4. Applications and validation

Griffin has been deployed in a number of full-core multiphysics simulation efforts for various reactor types, including Gas-Cooled (GC)-PBR, generic Fluoride salt-cooled High-temperature Reactor (gFHR), SFR, Lead-cooled Fast Reactor (LFR), Molten-Salt Reactor Experiment (MSRE), Molten Salt Fast Reactor (MSFR), generic Heat Pipe Micro Reactor (gHPMR), TREAT, High-Temperature engineering Test Reactor (HTTR), and Nuclear Thermal Propulsion (NTP), among others. A selected list of publications is included in Table 3 and encompasses a number of reactor designs and facilities in operation. As shown in this table, Griffin has recently been used to model a number of reactor designs for the Nuclear Regulatory Commission (NRC) Office of Nuclear Regulatory Research and the Advanced Reactor Technology (ART) program.

Griffin can model both gas- and fluoride-cooled high-temperature PBRs (Jaradat et al., 2023; Prince et al., 2023; Ortensi et al., 2023a). These models include equilibrium core multiphysics by coupling the Griffin depletion and core flux solutions with the MOOSE-based Pronghorn and BISON applications. For PBRs, Pronghorn solves the porous medium equations to obtain fluid and pebble surface temperature distributions and BISON solves the heat transfer to obtain pebble- and TRISO-level fuel and moderator temperatures. The gPBR-200 equilibrium core model results are shown in Fig. 17. The streamline method decay heat modeling for PBRs have successfully been performed using the pseudo-isotope feature in Griffin, described in Section 3.2.3. The results from Griffin PBR simulations have been compared to published work performed with the V.S.O.P. Teuchert et al. (1980) code for gas-cooled reactors and with Serpent for fluoride-cooled reactors. It is noted that in these studies the cross section libraries are prepared with the DRAGON (Marleau et al., 2011) lattice physics code using a two-step process. The Griffin team is currently deploying cross section methodologies for PBRs that will significantly improve its predictive capabilities. Nevertheless, the preliminary results show good agreement with established codes. For the HTR-PM design comparisons in Schunert et al. (2023), Griffin predicts a k-effective of 1.00150, where the reference value is assumed to be in a critical configuration since it is not published in the open literature. The power peaking results are 2.040 and 2.035 for V.S.O.P and Griffin, respectively. In addition, a comparison of the 2D power distribution can be found in Schunert et al. (2023). For the fluoride-cooled reactors, preliminary comparisons in the gFHR report (Ortensi et al., 2023a) show results from diffusion, SN transport and Monte Carlo calculations. It is noted that this work focused on code-coupling rather than reactor physics aspects alone. The discharge burnup is under-predicted in Griffin by 10.7% (diffusion) and 7.4% (S12), which can be attributed to the cross-section preparation and choice of group structure. The radial and axial power profiles are consistent with the Serpent-based reference solution. The report of gFHR (Ortensi et al., 2023a) also includes a comparison of isotopic compositions, reactivity coefficient, and kinetic parameters, where Griffin predicts a mean generation time of 3.75×10^{-4}s compared to the reference value of 3.86×10^{-4}s, and an effective delayed neutron fraction of 0.00536 compared to the reference value of 0.00539.

Griffin has been deployed for modeling fast reactor 3D dynamic problems with core thermal expansion (Terlizzi et al., 2024). In this case, the ring-heterogeneous spatial discretization with the SN transport solver in Griffin is coupled to BISON pin thermo-mechanics and SAM thermal fluids. A total of 61 flow channels and 60 pin models are used in this coupled model to simulate the unprotected loss-of-flow (ULOF) accident scenario. Comparisons to the Serpent (Leppänen,

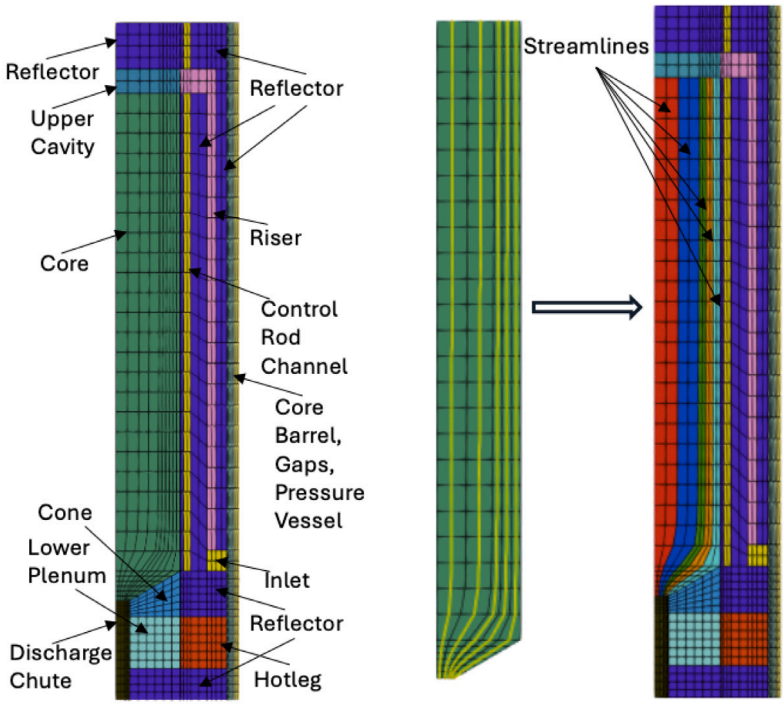


Fig. 16. Example of a typical mesh and geometry for a Griffin/Pronghorn model of a gas-cooled PBR in RZ-geometry (used in Ref. Prince et al. (2023)) is shown in 16(a), and a schematic illustrating the specification of streamlines is shown in 16(b) where the yellow lines passing through the core mesh are lines used to specify how the core mesh is grouped into five streamlines.

Table 3
Deployment of Griffin in reactor multiphysics analysis work. Partial validation work shown in **bold**.

Application	Description	Reference
GC-PBR	U.S. NRC reference plant model with transient multiphysics	Jaradat et al. (2023)
GC-PBR	Verification of the streamline method against Monte Carlo	de Oliveira et al. (2023)
GC-PBR	DLOFC transient multiphysics sensitivity	Prince et al. (2023)
gFHR	U.S. NRC reference plant model with transient multiphysics	Ortensi et al. (2023b)
SFR	U.S. NRC reference plant model with transient multiphysics	Terlizzi et al. (2024)
LFR	Hot channel factor analysis	Park et al. (2022c)
MSRE	**U.S. NRC reference plant model with transient multiphysics—Validation**	Jaradat and Ortensi (2023), Hartanto et al. (2023), Cao et al. (2023)
MSFR	U.S. NRC reference plant model with transient multiphysics	Terlizzi et al. (2021), Tano et al. (2022)
eVinci^{TM}	Westinghouse multiphysics analysis	Roskoff et al. (2022b,a)
gHPMR	U.S. NRC reference plant model with transient multiphysics	Ortensi et al. (2024)
other microReactors	Multiphysics Analysis of Load Following and Safety Transients for MicroReactors	Stauff et al. (2022, 2023), Abdelhameed et al. (2023), Terlizzi and Labouré (2023), Matthews et al. (2021)
SiMBA	Preparation of input data for MELCOR mechanistic source term accident analysis	Behne et al. (2023)
KRUSTY	Heat-pipe-cooled kilopower test reactor	Cao et al. (2024)
NTP	Multiphysics model of a nuclear thermal propulsion startup and shutdown	Labouré et al. (2023b)
TREAT	**Transient Reactor Test facility transient experiments—Validation**	Ortensi et al. (2021), Baker et al. (2016)
HTTR	**LOF transient experiments—Validation**	Labouré et al. (2023a)

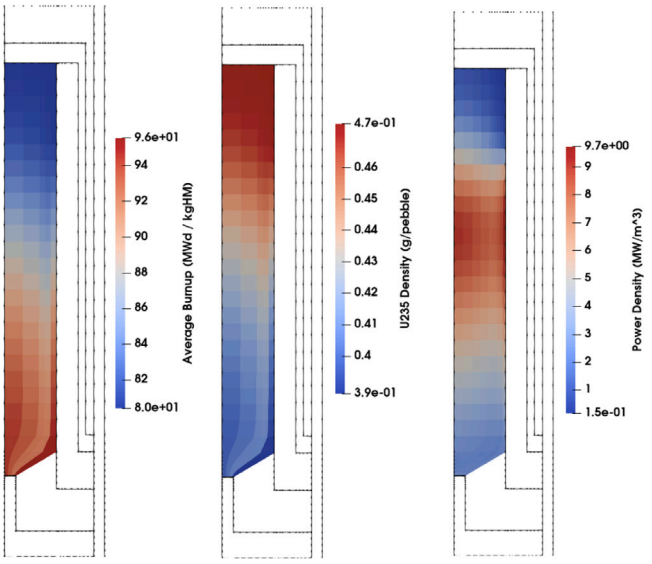


Fig. 17. Sample gPBR-200 (similar to the model used in Ref. Prince et al. (2023)) results showing power density, U235 concentration, and average burnup for the nominal equilibrium-core model.

2015) Monte Carlo code can be found in Terlizzi et al. (2024). They include 2D/3D assemblies and core configurations. We show that Griffin can predict the k-effective within 100 pcm of the reference Monte Carlo model for the full core using a ring-heterogeneous approach with MC^2-3-based (Lee and Yang, 2017) cross sections condensed to 9 energy groups. The ring-wise fission reaction rates for the 3D full core result in a RMS error of 1.31% and a maximum error of 3.72%, whereas the ring-wise absorption reaction rates result in a RMS error of 1.95% and a maximum error of 6.10%. The coupled steady-state results for the ABTR power, temperature and axial displacement fields are included in Fig. 18. During the ULOF transient, as shown in Fig. 19, the reduction in the sodium pump head leads to increased fuel and sodium temperatures that produce negative feedback reactivity, primarily from the radial displacement of the hex pad. It should be noted that Griffin solutions for the 3D ABTR core with homogenized assemblies (Lee et al., 2023) have been well-verified against Serpent Monte Carlo solutions, including eigenvalue, assembly powers, pin powers, and axial powers.

Another example is modeling heat-pipe-cooled microreactors. Fig. 20 includes the steady-state coupled solutions for the generic heat-pipe-cooled microreactor, a Westinghouse eVinci™-like model based on open literature (Ortensi et al., 2024) developed for the U.S. NRC. Fig. 21 shows results for a loss-of-heat-sink transient where the condenser still removes heat from the active core region. The microreactor exhibits early re-criticality oscillations due to the cooling effect from the heat pipes and the perfect thermal contact assumed between the fuel and monolith. In this case, the reactor will take a significant amount of time to reach equilibrium again.

The Griffin application has been validated for temperature-limited transients of the TREAT facility at INL. The SPH method in Griffin reproduces the reference solutions from the Serpent Monte Carlo code, but there are current limitations in the prediction of the kinetic parameters with SPH that require the adjustment to the control rod movement to match the reactor period. However, Griffin is able to predict the reactor power for a number of transients. The reactor power trace for a 1.5% $\Delta k/k$ reactivity insertion transient is shown in Fig. 22. The figure compares measurements from various detector channels with Griffin predictions using base cross-section data from ENDF/B-VII.1 and ENDF/B-VIII.0. The reported peak powers from the various detectors range between 524 and 593 MW, whereas the predicted peak powers from Griffin are 582 and 552 MW for ENDF/B-VII.1 and ENDF/B-VIII.0, respectively. The pulse width calculated by Griffin is 0.419 s in both cases, while the calculated values from the detector power traces range from 0.435 to 0.438 s. This validation includes spatial dynamics methods to model both the core and experiments. For the experiments, Griffin can predict accurate integral energy deposition within 5% of the measurements.

The Griffin validation for molten-salt reactors relies on the MSRE data set. The MSRE report (Jaradat and Ortensi, 2023) includes comparisons to the OpenMC (Romano et al., 2015) Monte Carlo code. Core eigenvalues are predicted within 150 pcm of the reference. Fuel and graphite temperature reactivity coefficients are 4.9 and −3.9% of the reference. The kinetics parameters for the MSRE stationary and flowing fuels are included in Table 9 of the MSRE report (Jaradat and Ortensi, 2023). The effective delayed neutron fraction computed with Griffin are 673 and 429 pcm for stationary and flowing fuel, respectively. Thus, the reactivity loss due to moving fuel is 244 pcm. The MSRE measured benchmark values from Haubenreich and Engel (1970) are 670 pcm for stationary fuel and approximately 240 pcm for reactivity loss due to moving fuel. The proper coupling between neutronics and thermal fluid fields is validated through the MSRE pump start-up and coast-down tests modeled with Griffin and SAM. Fig. 23 shows good agreement between the reactivity losses, from delayed neutrons decaying outside the active core region, computed between Griffin and the experiment.

Finally, The validation of Griffin for high-temperature reactors and some microreactor types includes the data set from the HTTR facility in Japan. The Griffin model was benchmarked against reference Serpent Monte Carlo calculations (Laboure et al., 2018). The SPH method shows very good agreement with the reference for both eigenvalue and power distributions, as shown in Tables 8, 9, 14, and 15 (Laboure et al., 2018). Eigenvalues match exactly the Monte Carlo reference, and the maximum error in the power distribution is less than 0.5%. Results for the HTTR 9 MWt loss-of-forced-coolant accident are included in Fig. 24. In this case, Griffin was coupled with BISON (Williamson et al., 2012) and THM (Hansel et al., 2024). The experimental data is not included here but will be presented in future publications for the 9 and 30 MWt LOFC transients.

5. Summary and future work

Griffin is the NEAMS reactor physics code that supports the analyses of various advanced reactor designs with deterministic transport methods. These advanced reactor designs often demand modeling capabilities with flexible geometry descriptions, multiphysics coupling, and support for features, such as TRISO fuel, control drums, etc., that are not supported by traditional analysis codes. The modular design in Griffin, especially its MOOSE-framework-based design, makes it able to meet challenges for modeling advanced reactor designs and adapt flexibly to accommodate new features. Griffin uses MOOSE/libMesh to support various finite element methods with unstructured mesh for radiation transport. It offers a DFEM-SN solver for heterogeneous models and HFEM-PN and various diffusion solvers for homogeneous models. These features make Griffin not only suitable for fast routine design calculations, where performance is the main factor, but also for reference calculations, where modeling accuracy is the primary consideration. Griffin can be seamlessly integrated with the MOOSE stochastic tools and optimization modules to generate reduced-order models that can be used for optimization calculations and for operational applications, such as core monitoring, operation support, etc. ISOXML, an object-oriented nuclear data toolkit in Griffin, is developed for multigroup cross-section management and is extended to support on-the-fly cross-section evaluations for thermal reactors, including PBRs. ISOXML provides capabilities to generate a simplified DTL from ENDF data, allowing for modeling neutron transport and decay heat accurately. Both the streamline-based method and PTT are available for PBR analyses. MC^2-3 can be optionally linked with

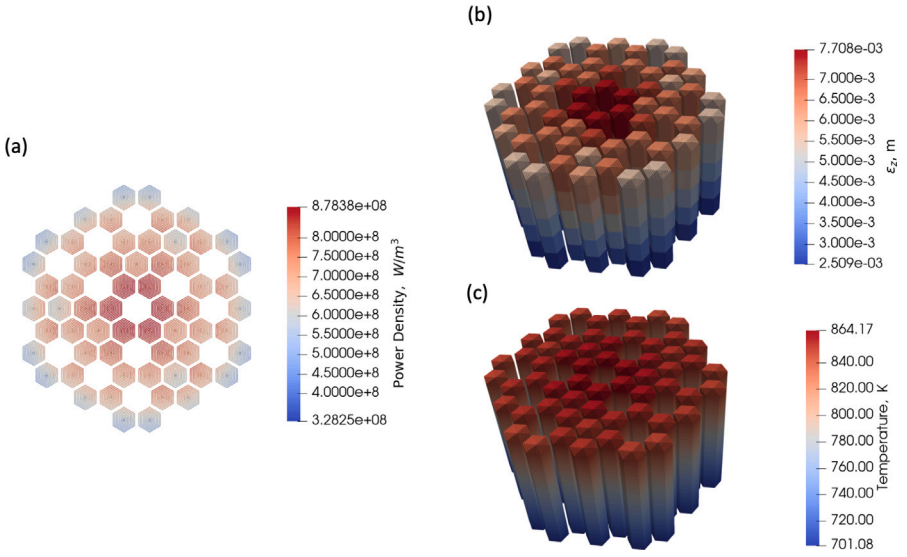
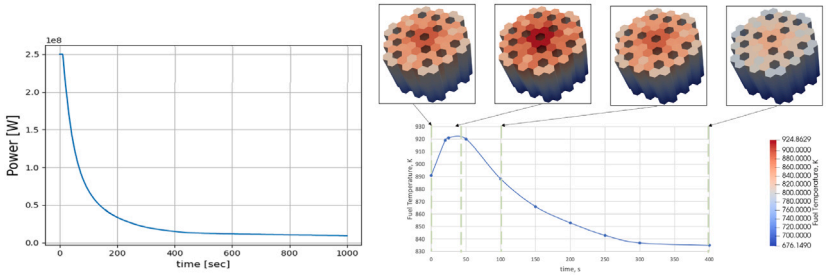
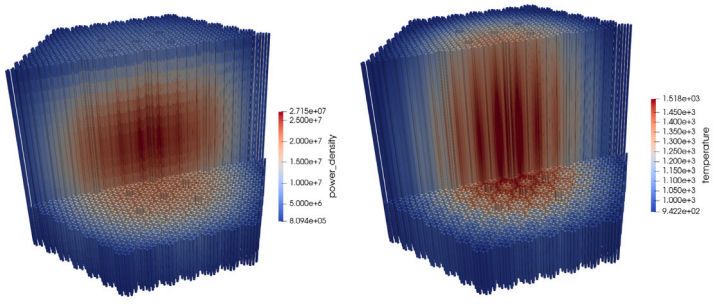


Fig. 18. (a) Power density for the fuel midplane, (b) 3D fuel temperature distribution, and (c) 3D fuel axial displacement (Terlizzi et al., 2024).



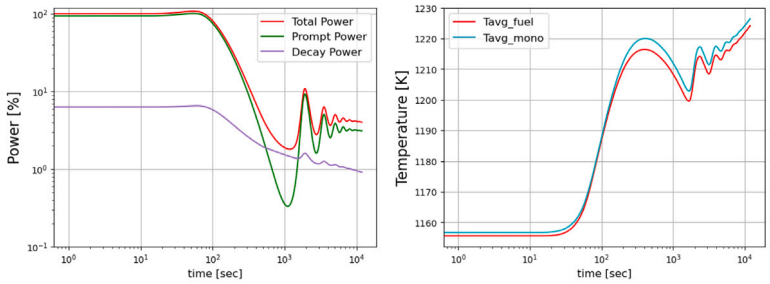
(a) Power (b) Temperature [K]

Fig. 19. ABTR unprotected loss-of-flow transient (Terlizzi et al., 2024).



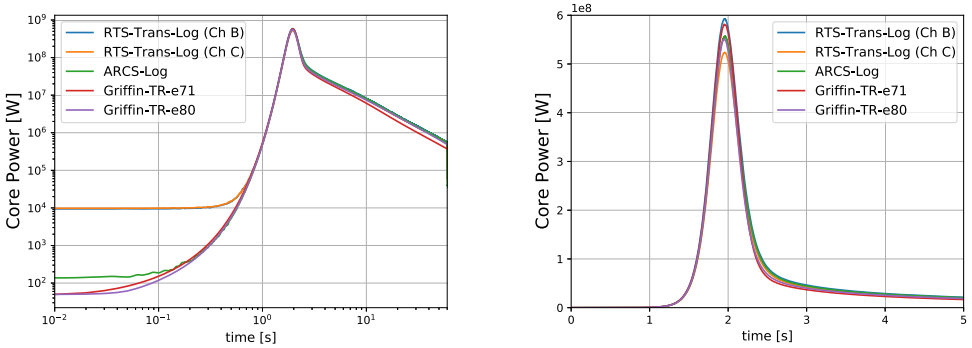
(a) Power density [W/m^3] (b) Temperature [K]

Fig. 20. gHPMR solutions for power and temperature with SN transport (Ortensi et al., 2024).



(a) Power density [W/m^3] (b) Temperature [K]

Fig. 21. gHPMR loss-of-heat-sink transient with SN transport (Ortensi et al., 2024).



(a) Log (b) Linear

Fig. 22. Core power for a temperature-limited transient (Ortensi et al., 2020).

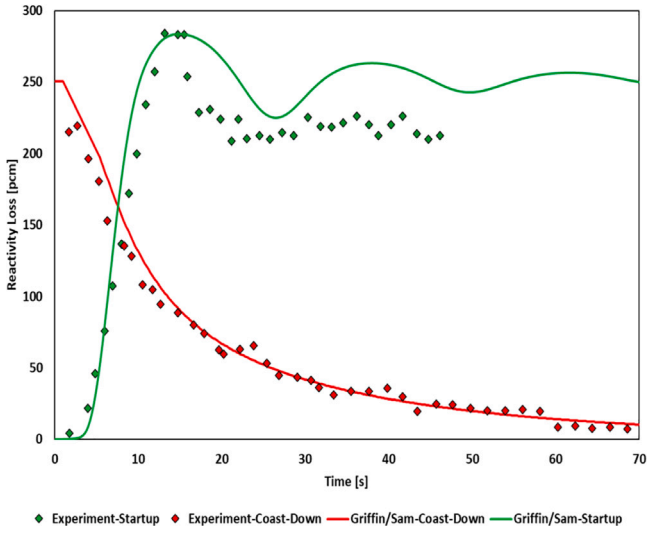


Fig. 23. Comparison calculated reactivity losses by Griffin/SAM and measured values data during pump start-up and coast-down tests (Jaradat and Ortensi, 2023).

Griffin to support fast-reactor analyses, including multiphysics transient simulations with Griffin. The depletion capability is flexible to support both heterogeneous and homogeneous calculations with fuel shuffling and multiple cycles for fixed solid fuel designs. Depletion is supported for PBRs where fuel pebbles are continuously recirculating in the core and for liquid-fueled reactors. Griffin can couple other physics either through MOOSE physics modules or other MOOSE-based applications, such as Pronghorn, SAM, Sockeye, etc., for modeling fuel performance, solid mechanics, fluid dynamics, and primary and secondary loops. Variables of all physics including neutron fluxes, multigroup cross sections, evaluated in a Griffin model, are exposed to users to allow flexible editing and post-processing. Griffin development follows the ASME NQA-1 SQA procedure, with which code documentation and review are enforced during code development. This not only produces a high-quality code but also ensures sustainable long-term development. In summary, Griffin is a unique reactor physics application designed to be flexible, compatible, extensible, and robust to meet modeling challenges from future-generation reactors.

Cross-section generation is one of the major areas for future development. We will enhance the NJOY module integration for multigroup library processing. Perturbation capabilities will be enhanced for a better user interface and optimization module integration. The performance of various models will be carefully scrutinized to minimize the memory and computing time requirements. The workflow and user interface of various models should be simplified into a special reactor analysis input syntax so that users can stay focused on reactor design and operation.

Contributions

The base technology in Griffin is from the Rattlesnake (Wang et al., 2021b), MAMMOTH (DeHart et al., 2019), the PROTEUS cross section API (CSAPI) (Lee and Jung, 2018b), and MC^2-3 development efforts. The contributions to that base technology are included in those publications.

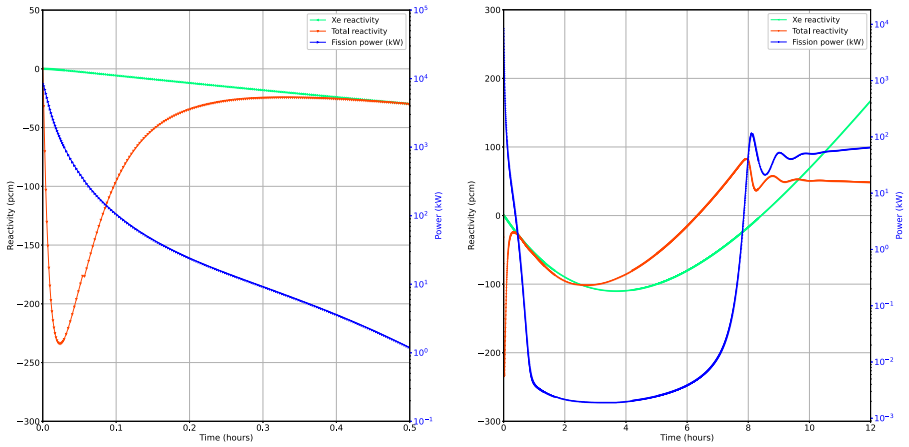
Yaqi Wang is the technical lead of the Griffin development effort. He led the software design and review, implemented the depletion capability, and added documentation for HFEM-PN. Zachary M. Prince implemented diffusion acceleration for the DFEM-SN scheme and IQS and was in charge of the MOOSE stochastic tools module integration. Hansol Park added the double heterogeneity treatment and online cross-section capability to SSAPI for PBR analyses. Olin W. Calvin worked on the decay transmutation library and Bateman solvers in ISOXML. Namjae Choi implemented various DFEM-SN optimizations and the ring-heterogeneous SFR model. Yeon Sang Jung implemented the HFEM-PN and HFEM-diffusion schemes, fuel management, SS-API integration, and MC^2-3 integration. Sebastian Schunert added the streamline-based equilibrium core calculation for PBRs. Shikhar Kumar implemented the pin-power reconstruction capability and worked on the MOOSE reactor module integration. Joshua T. Hanophy added the parallel transport sweeper and streamline-based running-in calculation for PBRs. Vincent M. Labouré implemented the homogenization equivalence with the multi-scheme and the xenon transient capability. Changho Lee worked on integrating MC^2-3 for the SFR workflow and led Griffin applications at ANL as well as verification in transport solvers, pin power reconstruction, cross sections, and the SFR workflow. Javier Ortensi is the lead of the Griffin team, including modelists, at INL. He led the PyNJOY/NJOY integration and various Griffin applications and validation at INL. Logan H. Harbour implemented the task scheduler for the transport sweeper and the sweep solver of the streamline-based PBR analysis interfacing MOOSE ray-tracing module. Jackson R. Harter added phonon and electron transport in the radiation transport module. All authors participated in the Griffin code review and maintenance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.



(a) First 1.5 hours (b) First 16 hours

Fig. 24. Evolution of the xenon and total reactivity for the 9 MWt LOFC transient (Labouré et al., 2022a).

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